Introduction to Analysis I

Lecture notes, Chapters 1–5 and Appendix

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Chapter 1. Real Numbers and Limits of Sequences

Analysis rests on a precise description of the real numbers. We take $\mathbb{R}$ as given by the field axioms, the order axioms and the completeness axiom, and derive everything else from them; Appendix A constructs a field satisfying the axioms. Book numbers are given in brackets.

1.1 Field and order axioms

Motivation. The algebraic and order rules of $\mathbb{R}$ are shared by $\mathbb{Q}$, so they cannot be what makes analysis work; still, every inequality manipulation later in the course reduces to them. We list them once and derive the familiar consequences, notably the triangle inequality.

Definition (Ordered field)

A **field** is a set F with operations $+$ and $\cdot$, both associative and commutative, with $\cdot$ distributive over $+$, identities $0 \neq 1$, additive inverses $-a$, and multiplicative inverses a^{-1} for $a \neq 0$ (axioms (F1)–(F9)). A field is **ordered** if there is a subset P (the positive elements) with (O1) $a, b \in P \Rightarrow a+b \in P$; (O2) $a, b \in P \Rightarrow ab \in P$; (O3) for each a exactly one of $a \in P$, $a = 0$, $-a \in P$ holds. We write $a < b$ (or $b > a$) if $b - a \in P$, and $a \leq b$ if $a < b$ or $a = b$.

Proposition 1.1 (1.1.1)

In a field:
(a) $-(-a) = a$, and $(a^{-1})^{-1} = a$ for $a \neq 0$;
(b) $a + b = a + c \Rightarrow b = c$, and $ab = ac$, $a \neq 0 \Rightarrow b = c$;
(c) $ab = 0 \iff a = 0$ or $b = 0$;
(d) $a \cdot 0 = 0$, $(-a)b = -(ab)$, $(-a)(-b) = ab$;
(e) the identities and inverses are unique.

Proof. (e) If $0, 0'$ are additive identities, $0 = 0 + 0' = 0'$; if b, b' are inverses of a, $b = b + (a + b') = (b + a) + b' = b'$; likewise for $\cdot$.

(b) Add $-a$ to both sides (multiply by a^{-1}).

(a) a and $-(-a)$ are both additive inverses of $-a$; use (e).

(d) $a \cdot 0 + a \cdot 0 = a(0 + 0) = a \cdot 0 = a \cdot 0 + 0$, so $a \cdot 0 = 0$ by (b); $ab + (-a)b = (a - a)b = 0$ gives $(-a)b = -(ab)$, and then $(-a)(-b) = -(a(-b)) = -(-(ab)) = ab$.

(c) If $ab = 0$ and $a \neq 0$, then $b = a^{-1}ab = 0$. ∎

Proposition 1.2 (1.1.2)

In an ordered field:
(a) $a \leq b$ and $b \leq a$ imply $a = b$;
(b) $a < b$, $b < c$ imply $a < c$;
(c) $a + b < a + c \iff b < c$;
(d) if $a > 0$ then $b < c \iff ab < ac$, and if $a < 0$ then $b < c \iff ab > ac$;
(e) $a^2 > 0$ for $a \neq 0$, in particular $1 > 0$;
(f) $a > 0 \Rightarrow a^{-1} > 0$, and $0 < a < b \Rightarrow 0 < b^{-1} < a^{-1}$;
(g) $a < b$, $c < d \Rightarrow a + c < b + d$;
(h) for $a, b \geq 0$: $a < b \iff a^2 < b^2$, and $a^2 = b^2 \Rightarrow a = b$.

Proof. (a) If $a \neq b$, both $b-a$ and $a-b=-(b-a)$ would be in P, against (O3).

(b) $c-a=(c-b)+(b-a)\in P$.

(c) $(a+c)-(a+b)=c-b$.

(d) $ac-ab=a(c-b)$; for $a>0$ use (O2) and trichotomy, for $a<0$ note $-a>0$.

(e) If $a>0$, $a^2\in P$; if $a<0$, $a^2=(-a)^2\in P$; and $1=1^2$.

(f) If $a^{-1}\leq 0$ then $1=aa^{-1}\leq 0$; multiply $a<b$ by $a^{-1}b^{-1}>0$.

(g) $(b+d)-(a+c)=(b-a)+(d-c)$.

(h) $b^2-a^2=(b-a)(b+a)$ and $b+a>0$ when $a,b\geq 0$ are not both 0. ■

$\mathbb{Q}$ and $\mathbb{R}$ are ordered fields. $\mathbb{C}$ is not: in an ordered field $-1<0<1=1^2$, but $i^2=-1$ would have to be positive. No finite field is ordered, since $1, 1+1, 1+1+1, \ldots$ are strictly increasing.

The **absolute value** is $|a|=a$ for $a\geq 0$ and $|a|=-a$ for $a<0$.

Proposition 1.3 (1.1.3)

(a) $|a|\geq 0$, and $|a|=0 \iff a=0$;
(b) $|ab|=|a||b|$;
(c) for $b\geq 0$, $|a|\leq b \iff -b\leq a\leq b$;
(d) $|a+b|\leq |a|+|b|$ (triangle inequality);
(e) $||a|-|b||\leq |a-b|$.

Proof. (a), (b) are case checks.

(c) If $a\geq 0$, $|a|\leq b \iff a\leq b$, and $-b\leq 0\leq a$ automatically; similarly for $a<0$.

(d) From $-|a|\leq a\leq |a|$ and the same for b, $-(|a|+|b|)\leq a+b\leq |a|+|b|$; apply (c).

(e) $|a|=|(a-b)+b|\leq |a-b|+|b|$, and symmetrically. ■

1.2 The completeness axiom

Motivation. $\mathbb{Q}$ has a "hole" at $\sqrt{2}$: the set $\{x\in\mathbb{Q}: x^2<2\}$ is bounded but has no least upper bound in $\mathbb{Q}$. The completeness axiom says $\mathbb{R}$ has no such holes, and it is the single property behind every existence theorem in analysis: limits of monotone sequences, intermediate values, maxima of continuous functions.

$S\subset\mathbb{R}$ is **bounded above** if it has an **upper bound** u ($s\leq u$ for all $s\in S$). A **least upper bound** (**supremum**) $\sup S$ is an upper bound smaller than every other upper bound; it is unique if it exists. Lower bounds and $\inf S$ are defined dually.

Definition (Completeness axiom)

(C) Every nonempty subset of $\mathbb{R}$ that is bounded above has a least upper bound.

Proposition 1.4 (1.2.1)

In an ordered field the following are equivalent:
(a) every nonempty set bounded above has a supremum;
(b) every nonempty set bounded below has an infimum.

Proof. (a)⇒(b): let $S \neq \emptyset$ be bounded below and L the set of lower bounds of S; $L \neq \emptyset$, and every $s \in S$ is an upper bound of L. Let $a = \sup L$. Each $s \in S$ is an upper bound of L, so $a \leq s$: $a \in L$. If $b \in L$ then $b \leq a$. So $a = \inf S$.

(b)⇒(a) is symmetric, or apply (b) to $-S = \{-s\}$. ■

A useful reformulation: $u = \sup S$ iff u is an upper bound and for every $\epsilon > 0$ there is $s \in S$ with $s > u - \epsilon$.

Proposition 1.5 (1.2.2)

$\mathbb{N}$ is not bounded above.

Proof. Otherwise $a = \sup \mathbb{N}$ exists; $a - 1$ is not an upper bound, so $a - 1 < n$ for some $n \in \mathbb{N}$, and then $n + 1 > a$ with $n + 1 \in \mathbb{N}$, a contradiction. ■

Corollary 1.6 (1.2.3)

$\inf\{1/n : n \in \mathbb{N}\} = 0$; that is, for every $\epsilon > 0$ there is $n \in \mathbb{N}$ with $1/n < \epsilon$.

Proof. 0 is a lower bound. If $\epsilon > 0$ were a lower bound, then $n \leq 1/\epsilon$ for all n, contradicting Proposition 1.5. ■

Corollary 1.7 (1.2.4, Archimedean property)

For $a > 0$ and $b \in \mathbb{R}$ there is $n \in \mathbb{N}$ with $na > b$.

Proof. By Proposition 1.5 choose $n > b/a$. ■

Theorem 1.8 (1.2.5, density of the rationals)

If $a < b$ there is $r \in \mathbb{Q}$ with $a < r < b$.

Proof. First let $0 \leq a < b$. Choose n with $1/n < b - a$ (Corollary 1.6). The set $\{m \in \mathbb{N} : m/n > a\}$ is nonempty (Corollary 1.7); let m be its least element. Then $(m-1)/n \leq a$ (if $m = 1$ this holds as $a \geq 0$), so $a < m/n \leq a + 1/n < b$. If $a < 0$, choose $k \in \mathbb{N}$ with $k > -a$, find r between $a + k \geq 0$ and $b + k$, and take $r - k$. ■

The irrationals are also dense: apply the theorem to $a/\sqrt{2} < b/\sqrt{2}$ (using Proposition 1.9).

Proposition 1.9 (1.2.6)

For each $a > 0$ there is a unique $b > 0$ with $b^2 = a$.

Proof. Uniqueness: Proposition 1.2(h).

Existence: $S = \{x > 0 : x^2 < a\}$ is nonempty (it contains $t = a/(1 + a)$, since $t < 1$ and $t < a$ give $t^2 < t < a$) and bounded above by $1 + a$ (if $x > 1 + a$ then $x^2 > x > a$). Let $b = \sup S > 0$. If $b^2 < a$, for $0 < h < 1$ we have $(b+h)^2 \leq b^2 + h(2b+1) < a$ once $h < (a - b^2)/(2b+1)$, so $b + h \in S$, a contradiction. If $b^2 > a$, for $0 < h < b$, $(b - h)^2 > b^2 - 2bh > a$ once $h < (b^2 - a)/2b$; then $b - h$ is an upper bound of S (if $x \geq b - h$ then $x^2 \geq (b - h)^2 > a$), a contradiction. So $b^2 = a$. ■

In particular $\mathbb{Q}$ does not satisfy (C): $\{x \in \mathbb{Q} : x > 0, x^2 < 2\}$ has no rational supremum, as $\sqrt{2}$ is irrational.

Theorem 1.10 (1.2.7, uniqueness of the real field)

Let F and G be ordered fields satisfying the completeness axiom. There is a bijection $f : F \to G$ with $f(a+b) = f(a) + f(b)$, $f(ab) = f(a)f(b)$, and $a < b \iff f(a) < f(b)$.

Proof. Each ordered field contains a copy of $\mathbb{Q}$: $n \mapsto 1 + \cdots + 1$, extended to $\mathbb{Z}$ and to quotients; this map $\mathbb{Q} \to F$ preserves operations and order (Proposition 1.2). Let $\phi : \mathbb{Q}_F \to \mathbb{Q}_G$ be the resulting isomorphism of the rational subfields. Both F and G are Archimedean (Corollary 1.7), so the rationals are dense in each (Theorem 1.8). Define

$$f(x) = \sup\{\phi(r) : r \in \mathbb{Q}_F,\ r < x\}.$$

The set is nonempty and bounded above (by $\phi(s)$ for any rational $s > x$), so $f(x)$ exists. Order: if $x < y$, pick rationals $x < r < s < y$; then $f(x) \le \phi(r) < \phi(s) \le f(y)$. Also $f(r) = \phi(r)$ for rational r (density). So f is strictly increasing, hence injective. Surjective: for $z \in G$ let $x = \sup\{r \in \mathbb{Q}_F : \phi(r) < z\}$; density in G and strict monotonicity show $f(x) = z$. Additivity: for rationals $r < x$, $s < y$ we have $r + s < x + y$, so $\phi(r) + \phi(s) \le f(x + y)$, and taking suprema $f(x) + f(y) \le f(x + y)$; applying this to $-x, -y$ (note $f(-x) = -f(x)$, from the definition and density) gives equality. Multiplicativity for $x, y > 0$: products rs of positive rationals $r < x$, $s < y$ are exactly cofinal among rationals below xy, giving $f(xy) = f(x)f(y)$; extend to all signs using $f(-x) = -f(x)$. ∎

So the axioms determine $\mathbb{R}$ up to isomorphism.

Proposition 1.11 (1.2.8)

For nonempty bounded $S, T \subset \mathbb{R}$, $\sup(S+T) = \sup S + \sup T$ and $\inf(S+T) = \inf S + \inf T$, where $S + T = \{s + t\}$. If S, T consist of positive numbers, $\sup(ST) = \sup S \cdot \sup T$ and $\inf(ST) = \inf S \cdot \inf T$.

Proof. $\sup S + \sup T$ is an upper bound of $S+T$. Given $\epsilon > 0$ choose $s > \sup S - \epsilon/2$, $t > \sup T - \epsilon/2$; then $s + t > \sup S + \sup T - \epsilon$. For products, $\sup S \sup T$ is an upper bound of ST; with $a = \sup S$, $b = \sup T$ and $0 < \delta < \min(a, b)$, choose $s > a - \delta$, $t > b - \delta$, so $st > (a-\delta)(b-\delta) = ab - \delta(a+b-\delta)$, which is arbitrarily close to ab. For infima, $\inf S \inf T$ is a lower bound of ST, and with $s < \inf S + \delta$, $t < \inf T + \delta$ we get $st < \inf S \inf T + \delta(\inf S + \inf T + \delta)$. ∎

1.3 Limits of sequences

Motivation. Limits are the basic operation of analysis, and the ε–N definition turns "gets arbitrarily close" into something we can prove things about. The limit laws below let us compute limits without returning to ε each time.

(x_n) **converges** to x, $\lim x_n = x$, if for every $\epsilon > 0$ there is N with $|x_n - x| < \epsilon$ for all $n \ge N$.

Proposition 1.12 (1.3.1)

Limits are unique.

Proof. If $x_n \to x$, $x_n \to y$ and $x \ne y$, take $\epsilon = |x - y|/2$ and $n \ge \max(N_1, N_2)$: $|x - y| \le |x - x_n| + |x_n - y| < 2\epsilon = |x - y|$. ∎

Proposition 1.13 (1.3.2)

Every convergent sequence is bounded.

Proof. If $x_n \to x$, choose N with $|x_n - x| < 1$ for $n \geq N$; then $|x_n| \leq \max\{|x_1|, \ldots, |x_{N-1}|, |x|+1\}$. ■

Theorem 1.14 (1.3.3, limit laws)

If $x_n \to x$ and $y_n \to y$, then (a) $x_n + y_n \to x + y$;
(b) $ax_n \to ax$;
(c) $x_n y_n \to xy$;
(d) if $y \neq 0$, then $y_n \neq 0$ for large n and $x_n/y_n \to x/y$;
(e) if $x_n \leq y_n$ for all large n, then $x \leq y$;
(f) (squeeze) if $x_n \leq z_n \leq y_n$ and $x = y$, then $z_n \to x$.

Proof. (a) $|(x_n + y_n) - (x + y)| \leq |x_n - x| + |y_n - y|$; use $\epsilon/2$ for each.

(b) is a special case of (c).

(c) With M bounding $|x_n|$ (Proposition 1.13), $|x_n y_n - xy| \leq |x_n||y_n - y| + |y||x_n - x| \leq M|y_n - y| + |y||x_n - x|$.

(d) For large n, $|y_n| > |y|/2$, and $|1/y_n - 1/y| = |y - y_n|/(|y_n||y|) \leq 2|y_n - y|/|y|^2$; combine with (c).

(e) If $x > y$, take $\epsilon = (x - y)/2$; for large n, $x_n > x - \epsilon = y + \epsilon > y_n$.

(f) For large n, $x - \epsilon < x_n \leq z_n \leq y_n < x + \epsilon$. ■

Example 1.1

(a) $1/n \to 0$ by Corollary 1.6.
(b) For $|r| < 1$, $r^n \to 0$: write $1/|r| = 1 + h$, $h > 0$; then $(1 + h)^n \geq 1 + nh$ (Bernoulli), so $|r|^n \leq 1/(1 + nh)$.
(c) $n^{1/n} \to 1$: with $n^{1/n} = 1 + h_n$, $n \geq \binom{n}{2} h_n^2$, so $0 \leq h_n \leq \sqrt{2/(n-1)}$.

1.4 Monotone sequences, limsup and liminf

Motivation. To prove a sequence converges without knowing its limit, we need intrinsic criteria. Monotone bounded sequences converge (this is completeness), and lim sup and lim inf always exist, measuring how far a sequence is from converging.

Theorem 1.15 (1.4.1, monotone convergence)

A bounded monotone sequence of real numbers converges.

Proof. Let (x_n) be increasing and bounded, and $x = \sup\{x_n\}$ (axiom (C)). Given ϵ, some $x_N > x - \epsilon$, and then $x - \epsilon < x_N \leq x_n \leq x$ for $n \geq N$. For decreasing sequences use inf. ■

Corollary 1.16 (1.4.2)

A series of nonnegative terms whose partial sums are bounded converges.

Proof. The partial sums form a bounded increasing sequence. ■

Example 1.2

$(1+1/n)^n$ is increasing and bounded by 3 (binomial expansion), so it converges to a number e. $\sum 1/n^p$ converges for $p>1$: grouping terms in blocks $2^k \le n < 2^{k+1}$, the partial sums are at most $\sum_k 2^{k(1-p)} < \infty$. $\sum 1/n$ diverges: the block sums are at least $\frac{1}{2}$.

For a bounded sequence (x_n), the sequences $\sup_{k\ge n} x_k$ and $\inf_{k\ge n} x_k$ are decreasing and increasing, so they converge (Theorem 1.15); define

$$\limsup_n x_n = \lim_n \sup_{k\ge n} x_k = \inf_n \sup_{k\ge n} x_k, \qquad \liminf_n x_n = \lim_n \inf_{k\ge n} x_k = \sup_n \inf_{k\ge n} x_k.$$

Clearly $\liminf x_n \le \limsup x_n$, and $\liminf x_n = -\limsup(-x_n)$.

Proposition 1.17 (1.4.3)

For bounded (x_n), $a = \limsup x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n < a+\epsilon$ for all $n \ge N$, and (b) for every $\epsilon > 0$, $x_n > a-\epsilon$ for infinitely many n.

Proof. Let $s_n = \sup_{k\ge n} x_k \downarrow a$.

(a) Choose N with $s_N < a+\epsilon$; then $x_n \le s_N$ for $n \ge N$.

(b) For each N, $s_N \ge a > a-\epsilon$, so some $k \ge N$ has $x_k > a-\epsilon$. Conversely, if a satisfies (a) and (b), then by (a) $s_n \le a+\epsilon$ for large n, so $\limsup \le a+\epsilon$; by (b) $s_n > a-\epsilon$ for all n, so $\limsup \ge a-\epsilon$. As ϵ is arbitrary, $\limsup x_n = a$. ■

Proposition 1.18 (1.4.4)

For bounded (x_n), $a = \liminf x_n$ iff (a) for every $\epsilon > 0$ there is N with $x_n > a-\epsilon$ for all $n \ge N$, and (b) for every $\epsilon > 0$, $x_n < a+\epsilon$ for infinitely many n.

Proof. Apply Proposition 1.17 to $(-x_n)$. ■

Theorem 1.19 (1.4.5)

If $x_n \to a$, then $\limsup x_n = \liminf x_n = a$. Conversely, if (x_n) is bounded and $\limsup x_n = \liminf x_n = a$, then $x_n \to a$.

Proof. If $x_n \to a$, then a satisfies conditions (a) and (b) of both propositions. Conversely, part (a) of Propositions 1.17 and 1.18 give $a-\epsilon < x_n < a+\epsilon$ for all large n. ■

Theorem 1.20 (1.4.6, nested interval theorem)

If $I_n = [a_n, b_n]$ are bounded closed intervals with $I_{n+1} \subset I_n$ for all n, then $\bigcap_n I_n \ne \emptyset$. If moreover $b_n - a_n \to 0$, the intersection is a single point.

Proof. (a_n) is increasing and bounded by b_1, so $a_n \to a = \sup a_n$; for all m, n, $a_n \le a_{\max(m,n)} \le b_{\max(m,n)} \le b_m$, so $a \le b_m$. Thus $a_m \le a \le b_m$ for all m. If x, y both lie in every I_n, $|x-y| \le b_n - a_n \to 0$. ■

Conversely, the nested interval theorem together with the Archimedean property implies the completeness axiom: for S bounded above, bisect an interval $[s_0, u_0]$ ($s_0 \in S$, u_0 an upper bound) repeatedly, keeping the half whose right end point is an upper bound and whose interval meets S; the common point of the intervals is $\sup S$. Similarly, Theorem 1.15 is equivalent to (C).

1.5 Countable and uncountable sets

Motivation. $\mathbb{Q}$ and $\mathbb{R}$ are both dense, but there are "more" reals than rationals. Countability gives a precise way to say this and is used, for example, to show that discontinuities of monotone functions are rare.

A set is **countable** if it is finite or in bijection with $\mathbb{N}$.

Proposition 1.21 (countable unions)

Subsets of countable sets are countable; $\mathbb{N}\times\mathbb{N}$ is countable; a countable union of countable sets is countable. Hence $\mathbb{Z}$ and $\mathbb{Q}$ are countable.

Proof. An infinite $A\subset\mathbb{N}$ is enumerated increasingly by recursion ($a_1=\min A$, $a_{k+1}=\min(A-\{a_1,\dots,a_k\})$). $(m,n)\mapsto 2^m3^n$ injects $\mathbb{N}\times\mathbb{N}$ into $\mathbb{N}$. If A_k are countable with surjections $f_k:\mathbb{N}\to A_k$ (for nonempty A_k), then $(k,n)\mapsto f_k(n)$ maps $\mathbb{N}\times\mathbb{N}$ onto $\bigcup A_k$, and a set that is the image of a countable set is countable (choose least preimages). $\mathbb{Q}$ is the image of $\mathbb{Z}\times\mathbb{N}$ under $(p,q)\mapsto p/q$. ■

Theorem 1.22 (Cantor)

$\mathbb{R}$ is uncountable.

Proof. Let $x_1,x_2,\dots$ be any sequence of reals. Choose a closed interval I_1 of positive length not containing x_1, then $I_2\subset I_1$ not containing x_2, and so on. By Theorem 1.20 some $x\in\bigcap I_n$, and $x\neq x_n$ for every n. ■

Consequently the irrational numbers are uncountable, and every interval of positive length is uncountable.

1.6 Selected exercises

Motivation. Three problems from the book's exercise set that use the tools of this chapter in an essential way: a tail estimate for a series, monotone convergence for a pair of coupled sequences, and countability.

Exercise 1.6.10 (e is irrational)

Let $s_n=\sum_{k=0}^{n}\frac{1}{k!}$.
(a) Show that (s_n) converges; call its limit e. (It agrees with $\lim(1+1/n)^n$ of Example 1.2, which is not needed here.)
(b) Show that $0<e-s_n<\dfrac{1}{n!\,n}$ for every $n\geq 1$.
(c) Show that e is irrational.

Solution.

(a) and (b).

Step 1. For $k\geq n+1$ we have $k!=(n+1)!\,(n+2)(n+3)\cdots k\geq(n+1)!\,(n+2)^{k-n-1}$. Hence for $m>n$,

$$0<s_m-s_n=\sum_{k=n+1}^{m}\frac{1}{k!}\leq\frac{1}{(n+1)!}\sum_{j=0}^{\infty}\frac{1}{(n+2)^j}=\frac{1}{(n+1)!}\cdot\frac{n+2}{n+1}.$$

Step 2. With $n=1$ this gives $s_m \le s_1 + \frac{1}{2}\cdot\frac{3}{2} = \frac{11}{4}$ for all m. So (s_m) is increasing and bounded, and converges by Theorem 1.15.

Step 3. Letting $m \to \infty$ in Step 1 (limits preserve $\le$), and using $s_m - s_n \ge s_{n+1} - s_n$ for $m > n$,

$$0 < \frac{1}{(n+1)!} \le e - s_n \le \frac{n+2}{(n+1)!\,(n+1)} = \frac{1}{n!\,n}\cdot\frac{n(n+2)}{(n+1)^2} < \frac{1}{n!\,n},$$

since $n(n+2) = (n+1)^2 - 1$.

(c)

Step 1. Suppose $e = p/q$ with $p, q \in \mathbb{N}$, $q \ge 1$ (note $e > s_1 = 2 > 0$). Then $q!\,e = p\,(q-1)!$ is an integer, and $q!\,s_q = \sum_{k=0}^{q} q!/k!$ is an integer, because $k!$ divides $q!$ for $k \le q$.

Step 2. So $q!(e - s_q)$ is an integer, while by (b) $0 < q!(e - s_q) < 1/q \le 1$. No integer lies strictly between 0 and 1, a contradiction. Hence $e \notin \mathbb{Q}$. ∎

Exercise 1.6.15 (Arithmetic–geometric mean)

Let $a_1 \ge b_1 > 0$ and define recursively

$$a_{n+1} = \frac{a_n + b_n}{2}, \qquad b_{n+1} = \sqrt{a_n b_n}.$$

Show that (a_n) and (b_n) converge to the same limit.

Solution. If $a_1 = b_1$, both sequences are constant. Let $a_1 > b_1$. Square roots exist by Proposition 1.9.

Step 1. For $x > y > 0$ we have $\frac{x+y}{2} > \sqrt{xy}$, because $\frac{x+y}{2} - \sqrt{xy} = \frac{1}{2}(\sqrt{x} - \sqrt{y})^2 > 0$. By induction, $a_n > b_n > 0$ for every n.

Step 2. Then $a_{n+1} = \frac{a_n + b_n}{2} < a_n$ and $b_{n+1} = \sqrt{a_n b_n} > \sqrt{b_n^2} = b_n$. So

$$b_1 < b_2 < \cdots < b_n < a_n < \cdots < a_2 < a_1.$$

(a_n) is decreasing and bounded below by b_1, and (b_n) is increasing and bounded above by a_1; by Theorem 1.15, $a_n \to L$ and $b_n \to M$.

Step 3. Since $b_{n+1} > b_n$,

$$0 < a_{n+1} - b_{n+1} < a_{n+1} - b_n = \frac{a_n - b_n}{2},$$

so $0 < a_n - b_n < (a_1 - b_1)/2^{n-1} \to 0$. By the limit laws (Theorem 1.14), $L - M = \lim(a_n - b_n) = 0$. ∎

Exercise 1.6.17 (Algebraic numbers)

A real number is **algebraic** if it is a root of a nonzero polynomial with integer coefficients, and **transcendental** otherwise.
(a) Show that the set of algebraic numbers is countable.
(b) Deduce that there are uncountably many transcendental numbers.

Solution.

(a)

Step 1. If A, B are countable, so is $A \times B = \bigcup_{a \in A} \{a\} \times B$, a countable union of countable sets (Proposition 1.21). By induction $\mathbb{Z}^{d+1}$ is countable for each $d \in \mathbb{N}$.

Step 2. Let P_d be the set of nonzero polynomials of degree at most d with integer coefficients. The map $c_0 + c_1x + \cdots + c_dx^d \mapsto (c_0, \ldots, c_d)$ injects P_d into $\mathbb{Z}^{d+1}$, so P_d is countable, and so is $P = \bigcup_{d\in\mathbb{N}} P_d$.

Step 3. A nonzero polynomial p of degree d has at most d real roots. Indeed, for $d = 0$ it has none; if $d \geq 1$ and $p(r) = 0$, division by $x - r$ gives $p(x) = (x - r)q(x) + p(r) = (x - r)q(x)$ with $\deg q = d - 1$, and a root $s \neq r$ of p is a root of q (Proposition 1.1(c)); apply induction to q.

Step 4. The set of algebraic numbers is $\bigcup_{p\in P}\{x \in \mathbb{R} : p(x) = 0\}$, a countable union of finite sets, hence countable by Proposition 1.21.

(b) Let T be the set of transcendental numbers. If T were countable, $\mathbb{R} = T \cup (\mathbb{R} - T)$ would be a union of two countable sets, hence countable, contradicting Theorem 1.22. So T is uncountable; in particular it is nonempty. ■

Chapter 2. Topology of Euclidean Space

2.1 Euclidean space

Motivation. To study functions of several variables we need distance in $\mathbb{R}^n$. The Cauchy–Schwarz inequality is what makes the Euclidean distance satisfy the triangle inequality.

$\mathbb{R}^n$ is the set of n-tuples $x = (x_1, \dots, x_n)$ with coordinatewise addition and scalar multiplication, the **inner product** $\langle x, y\rangle = \sum x_i y_i$, the **norm** $\|x\| = \langle x, x\rangle^{1/2}$ and the **distance** $\|x - y\|$.

Theorem 2.1 (2.1.1, Cauchy–Schwarz inequality)

$|\langle x, y\rangle| \le \|x\|\|y\|$ for all $x, y \in \mathbb{R}^n$, with equality iff x, y are linearly dependent. Consequently $\|x + y\| \le \|x\| + \|y\|$ and $\|x - z\| \le \|x - y\| + \|y - z\|$.

Proof. If $y = 0$ both sides vanish. Otherwise, for real t, $0 \le \|x - ty\|^2 = \|x\|^2 - 2t\langle x, y\rangle + t^2\|y\|^2$; take $t = \langle x, y\rangle / \|y\|^2$ to get $0 \le \|x\|^2 - \langle x, y\rangle^2/\|y\|^2$. Equality forces $x = ty$. Then $\|x + y\|^2 = \|x\|^2 + 2\langle x, y\rangle + \|y\|^2 \le (\|x\| + \|y\|)^2$. ∎

Also $\max_i |x_i| \le \|x\| \le \sqrt{n}\max_i |x_i|$, so a sequence in $\mathbb{R}^n$ converges iff each coordinate sequence converges.

2.2 Open and closed sets

Motivation. Many theorems need a domain without boundary points (to differentiate) or one containing all its limits (to guarantee maxima). Open and closed sets express these two requirements.

The **ϵ-neighbourhood** of x is $N(x, \epsilon) = \{y : \|y - x\| < \epsilon\}$. A point x is an **interior point** of A if some $N(x, \epsilon) \subset A$; A is **open** if all its points are interior. x is a **limit point** of A if every $N(x, \epsilon)$ contains a point of A other than x; A' denotes the set of limit points, and $\overline{A} = A \cup A'$ is the **closure**. A is **closed** if $\mathbb{R}^n - A$ is open. Points of $A - A'$ are **isolated points**. Every $N(x, \epsilon)$ is open, by the triangle inequality.

Proposition 2.2 (2.2.1)

(a) $\emptyset$ and $\mathbb{R}^n$ are open;
(b) unions of open sets are open;
(c) finite intersections of open sets are open. Dually, $\emptyset, \mathbb{R}^n$ are closed, intersections of closed sets are closed, and finite unions of closed sets are closed.

Proof. (a), (b) are immediate.

(c) If $x \in U_1 \cap \cdots \cap U_k$ with $N(x, \epsilon_i) \subset U_i$, then $N(x, \min \epsilon_i)$ lies in the intersection. The dual statements follow by De Morgan's laws. ∎

An infinite intersection of open sets need not be open: $\bigcap_n (-1/n, 1/n) = \{0\}$.

Proposition 2.3 (2.2.2)

For $A \subset \mathbb{R}^n$ and $x \in \mathbb{R}^n$:
(a) $x \in A'$ iff some sequence in $A - \{x\}$ converges to x;
(b) $x \in \overline{A}$ iff some sequence in A converges to x.

Proof. (a) If $x \in A'$, pick $x_k \in A \cap N(x, 1/k)$, $x_k \neq x$. Conversely, if $x_k \to x$ with $x_k \in A - \{x\}$, every $N(x, \epsilon)$ contains some x_k.

(b) If $x \in A$ use the constant sequence; otherwise $x \in A'$ and use (a). Conversely, if $x_k \to x$, $x_k \in A$, either $x = x_k$ for some k or the x_k lie in $A - \{x\}$. ■

Theorem 2.4 (2.2.3)

For $F \subset \mathbb{R}^n$ the following are equivalent:
(a) F is closed;
(b) $F' \subset F$;
(c) $F = \overline{F}$.

Proof. (a)⇒(b): if $x \notin F$, some $N(x, \epsilon)$ misses F, so $x \notin F'$. (b)⇒(c): $\overline{F} = F \cup F' = F$. (c)⇒(a): if $x \notin F$, then $x \notin F'$, so some $N(x, \epsilon)$ contains no point of F (other than x, which is not in F); thus $\mathbb{R}^n - F$ is open. ■

Corollary 2.5 (2.2.4)

A' is closed for every $A \subset \mathbb{R}^n$; hence $\overline{A}$ is closed, and it is the smallest closed set containing A.

Proof. Let $x \in (A')'$ and $\epsilon > 0$. Choose $y \in A'$ with $0 < \|y - x\| < \epsilon$, and $\delta = \min(\|y - x\|, \epsilon - \|y - x\|) > 0$. Then $N(y, \delta)$ contains a point $z \in A$ with $z \neq y$; $z \in N(x, \epsilon)$, and $z \neq x$ because $\|z - y\| < \|x - y\|$. So $x \in A'$. The same argument shows $(\overline{A})' \subset A'$: if $N(x, \epsilon)$ contains $y \in A \cup A'$ with $y \neq x$, either $y \in A$, or $y \in A'$ and a small neighbourhood of y inside $N(x, \epsilon)$ contains a point of A other than x. So $\overline{A}$ is closed by Theorem 2.4. If $F \supset A$ is closed, $F \supset F' \supset A'$, so $F \supset \overline{A}$. ■

For $X \subset \mathbb{R}^n$, a set $U \subset X$ is **open in** X (relatively open) if for each $x \in U$ there is $\epsilon > 0$ with $N(x, \epsilon) \cap X \subset U$.

Proposition 2.6 (2.2.5)

For $U \subset X \subset \mathbb{R}^n$ the following are equivalent:
(a) U is open in X;
(b) $U = V \cap X$ for some open $V \subset \mathbb{R}^n$.

Proof. (a)⇒(b): for $x \in U$ choose ϵ_x with $N(x, \epsilon_x) \cap X \subset U$, and let $V = \bigcup_x N(x, \epsilon_x)$. (b)⇒(a): if $x \in V \cap X$ and $N(x, \epsilon) \subset V$, then $N(x, \epsilon) \cap X \subset U$. ■

Similarly, $F \subset X$ is closed in X iff $F = E \cap X$ with E closed in $\mathbb{R}^n$.

2.3 The Bolzano–Weierstrass theorem and Cauchy sequences

Motivation. Every bounded sequence has a convergent subsequence. This is the main way to produce limits out of nothing, and it shows that a sequence converges exactly when its terms eventually stay close to each other (the Cauchy criterion), which can be checked without knowing the limit.

A **box** is $B = I_1 \times \cdots \times I_n$ with bounded closed intervals I_k.

Lemma 2.7 (2.3.1, nested boxes)

If $B_1 \supset B_2 \supset \cdots$ are boxes, then $\bigcap_m B_m \neq \emptyset$.

Proof. Write $B_m = I_1^m \times \cdots \times I_n^m$. For each k, the intervals I_k^m are nested, so contain a common point x_k (Theorem 1.20); then $(x_1, \ldots, x_n) \in \bigcap B_m$. ∎

Theorem 2.8 (2.3.2, Bolzano–Weierstrass)

Every bounded infinite set $A \subset \mathbb{R}^n$ has a limit point.

Proof. A lies in a box B_1. Divide B_1 into 2^n boxes by bisecting each side; one of them, B_2, contains infinitely many points of A. Repeating, we obtain nested boxes B_m whose sides have length 2^{1-m} times those of B_1, each containing infinitely many points of A. Let $x \in \bigcap B_m$ (Lemma 2.7). Any $N(x, \epsilon)$ contains B_m for large m (its diameter tends to 0), hence infinitely many points of A; so $x \in A'$. ∎

Proposition 2.9 (2.3.3)

If x is a limit point of A, then $A \cap N(x, \epsilon)$ is infinite for every $\epsilon > 0$.

Proof. If $A \cap N(x, \epsilon) - \{x\} = \{a_1, \ldots, a_k\}$, then $N(x, \delta)$ with $\delta = \min\|a_i - x\|$ contains no point of A other than x, a contradiction. ∎

Theorem 2.10 (2.3.4)

Every bounded sequence in $\mathbb{R}^n$ has a convergent subsequence.

Proof. If the range $\{x_k\}$ is finite, some value is taken infinitely often, giving a constant subsequence. Otherwise it is a bounded infinite set with a limit point x (Theorem 2.8). Each $N(x, 1/j)$ contains infinitely many terms (Proposition 2.9), so we can choose $k_1 < k_2 < \cdots$ with $\|x_{k_j} - x\| < 1/j$. ∎

For a bounded real sequence (a_n) let A be the set of limits of its convergent subsequences.

Theorem 2.11 (2.3.5)

$\limsup a_n = \max A$ and $\liminf a_n = \min A$.

Proof. Let $a = \limsup a_n$. By Proposition 1.17(b) we may choose $n_1 < n_2 < \cdots$ with $a_{n_k} > a - 1/k$, and by (a) of the same proposition $a_{n_k} < a + 1/k$ for large k; so $a_{n_k} \to a$ and $a \in A$. If $a_{n_k} \to b$, then $a_{n_k} < a + \epsilon$ for large k, so $b \leq a$. The $\liminf$ case is symmetric. ∎

A sequence is **Cauchy** if for every $\epsilon > 0$ there is N with $\|x_m - x_n\| < \epsilon$ for all $m, n \geq N$.

Proposition 2.12 (2.3.6)

Every convergent sequence is Cauchy, and every Cauchy sequence is bounded.

Proof. If $x_n \to x$, choose N with $\|x_n - x\| < \epsilon/2$ for $n \geq N$; then $\|x_m - x_n\| < \epsilon$. If (x_n) is Cauchy, choose N for $\epsilon = 1$; then $\|x_n\| \leq \max\{\|x_1\|, \ldots, \|x_N\|\} + 1$. ∎

Lemma 2.13 (2.3.7)

If a subsequence of a Cauchy sequence (x_n) converges to x, then (x_n) converges to x.

Proof. Given ϵ, choose N with $\|x_m - x_n\| < \epsilon/2$ for $m, n \geq N$, and k with $n_k \geq N$ and $\|x_{n_k} - x\| < \epsilon/2$. For $n \geq N$, $\|x_n - x\| \leq \|x_n - x_{n_k}\| + \|x_{n_k} - x\| < \epsilon$. ■

Theorem 2.14 (2.3.8, Cauchy criterion)

Every Cauchy sequence in $\mathbb{R}^n$ converges.

Proof. It is bounded (Proposition 2.12), so has a convergent subsequence (Theorem 2.10); apply Lemma 2.13. ■

Corollary 2.15 (2.3.9, Cauchy criterion for series)

$\sum a_n$ converges iff for every $\epsilon > 0$ there is N with $|a_{m+1} + \cdots + a_n| < \epsilon$ for all $n > m \geq N$.

Proof. Apply Theorem 2.14 to the partial sums s_n, since $s_n - s_m = a_{m+1} + \cdots + a_n$. ■

In particular, if $\sum a_n$ converges then $a_n \to 0$, and if $\sum |a_n|$ converges (**absolute convergence**) then $\sum a_n$ converges, because $|a_{m+1} + \cdots + a_n| \leq |a_{m+1}| + \cdots + |a_n|$.

2.4 Series

Motivation. Infinite sums are limits of partial sums, so convergence of series is a special case of convergence of sequences. The tests below (comparison, ratio, root, alternating) are practical forms of the Cauchy criterion, and absolute convergence explains when terms may be rearranged.

Proposition 2.16 (2.4.1, rearrangement of nonnegative series)

Let $a_n \geq 0$ and $\tau : \mathbb{N} \to \mathbb{N}$ a bijection. If $\sum a_n = s$, then $\sum a_{\tau(n)} = s$; if $\sum a_n$ diverges, so does $\sum a_{\tau(n)}$.

Proof. Each partial sum $\sum_{n \leq N} a_{\tau(n)}$ is at most $\sum_{n \leq M} a_n \leq s$ with $M = \max \tau(\{1, \ldots, N\})$, so $t = \sum a_{\tau(n)} \leq s$ by Corollary 1.16. Applying this to τ^{-1} gives $s \leq t$. The divergence statement follows by symmetry. ■

More generally, an absolutely convergent series may be rearranged without changing its sum: split a_n into positive and negative parts $a_n^{\pm} = \max(\pm a_n, 0)$, both summable, and apply the proposition to each. By contrast, a conditionally convergent series can be rearranged to have any sum (Riemann).

Theorem 2.17 (2.4.2, alternating series test)

If $x_1 \geq x_2 \geq \cdots \geq 0$ and $x_n \to 0$, then $\sum (-1)^{n+1} x_n$ converges, and its sum s satisfies $|s - s_n| \leq x_{n+1}$.

Proof. $s_{2m} = (x_1 - x_2) + \cdots + (x_{2m-1} - x_{2m})$ increases, $s_{2m+1} = x_1 - (x_2 - x_3) - \ldots$ decreases, and $s_{2m} \leq s_{2m+1}$. Both converge (Theorem 1.15), and $s_{2m+1} - s_{2m} = x_{2m+1} \to 0$, so they have the same limit s, which lies between consecutive partial sums. ■

Theorem 2.18 (2.4.3, comparison test)

If $0 \le a_n \le b_n$ and $\sum b_n$ converges, then $\sum a_n$ converges.

Proof. The partial sums of $\sum a_n$ are bounded by $\sum b_n$; use Corollary 1.16. ■

Proposition 2.19 (2.4.4, root test)

Let $\alpha = \limsup |a_n|^{1/n}$. If $\alpha < 1$, $\sum a_n$ converges absolutely; if $\alpha > 1$, $\sum a_n$ diverges.

Proof. If $\alpha < \beta < 1$, then $|a_n| < \beta^n$ for large n (Proposition 1.17), and $\sum \beta^n < \infty$. If $\alpha > 1$, $|a_n| > 1$ for infinitely many n, so $a_n \not\to 0$. ■

Proposition 2.20 (2.4.5, ratio test)

Let $a_n \neq 0$. If $\limsup |a_{n+1}/a_n| < 1$, then $\sum a_n$ converges absolutely; if $|a_{n+1}/a_n| \ge 1$ for all large n, then $\sum a_n$ diverges.

Proof. If $|a_{n+1}/a_n| \le \beta < 1$ for $n \ge N$, then $|a_n| \le |a_N|\beta^{n-N}$. In the second case $|a_n|$ is eventually nondecreasing and positive, so $a_n \not\to 0$. ■

Proposition 2.21 (2.4.6)

If $c_n > 0$, then $\liminf \frac{c_{n+1}}{c_n} \le \liminf c_n^{1/n} \le \limsup c_n^{1/n} \le \limsup \frac{c_{n+1}}{c_n}$.

Proof. Let $\beta = \limsup c_{n+1}/c_n$, assumed finite, and $\gamma > \beta$. For $n \ge N$, $c_{n+1} < \gamma c_n$, so $c_n < c_N \gamma^{-N} \gamma^n$ and $c_n^{1/n} < (c_N \gamma^{-N})^{1/n} \gamma \to \gamma$. Hence $\limsup c_n^{1/n} \le \gamma$ for all $\gamma > \beta$. The left inequality is analogous. ■

So the root test is stronger than the ratio test: whenever the ratio test decides, so does the root test.

2.5 Compact sets

Motivation. A closed bounded set in $\mathbb{R}^n$ has the property that every open cover has a finite subcover. This abstract property is what lets us pass from local estimates to global ones, and Heine–Borel shows it is the same as being closed and bounded.

An **open cover** of K is a family of open sets whose union contains K; K is **compact** if every open cover has a finite subcover.

Proposition 2.22 (2.5.1)

Every compact set is bounded and closed.

Proof. The sets $N(0,k)$ cover K; finitely many do, so $K \subset N(0,m)$. If $x \notin K$, the open sets $U_k = \{y : \|y - x\| > 1/k\}$ cover K; a finite subcover gives $K \subset U_m$, so $N(x, 1/m)$ misses K. Hence $\mathbb{R}^n - K$ is open. ■

Theorem 2.23 (2.5.2)

Every box $B = I_1 \times \cdots \times I_n$ is compact.

Proof. Suppose an open cover $\{U_i\}$ of B has no finite subcover. Bisecting all sides, one of the 2^n sub-boxes, B_2, is not covered by finitely many U_i; continuing gives nested boxes B_m of diameter $\operatorname{diam} B_1/2^{m-1}$ none finitely covered. Let $x \in \bigcap B_m$ (Lemma 2.7) and $x \in U_i$, $N(x,\epsilon) \subset U_i$. For large m, $B_m \subset N(x,\epsilon) \subset U_i$, a contradiction. ■

Proposition 2.24 (2.5.3)

A closed subset F of a compact set K is compact.

Proof. If $\{U_i\}$ covers F, then $\{U_i\} \cup \{\mathbb{R}^n - F\}$ covers K; a finite subcover, with $\mathbb{R}^n - F$ removed, covers F. ■

Theorem 2.25 (2.5.4, Heine–Borel)

For $K \subset \mathbb{R}^n$ the following are equivalent:
(a) K is compact;
(b) K is bounded and closed;
(c) every infinite subset of K has a limit point in K.

Proof. (a)⇒(b): Proposition 2.22. (b)⇒(a): K lies in a box, which is compact (Theorem 2.23); apply Proposition 2.24. (b)⇒(c): Bolzano–Weierstrass gives a limit point, which is in K as K is closed. (c)⇒(b): if K were unbounded, choose $x_k \in K$ with $\|x_k\| > k$; the infinite set $\{x_k\}$ has no limit point at all. If $x \in K'$, Proposition 2.9 gives distinct $x_k \in K$ with $x_k \to x$; the set $\{x_k\}$ has x as its only limit point, so $x \in K$ by (c), and K is closed. ■

So K is compact iff every sequence in K has a subsequence converging in K.

Theorem 2.26 (2.5.5, finite intersection property)

If $\{K_i : i \in I\}$ are compact sets such that $\bigcap_{i\in J} K_i \neq \emptyset$ for every finite $J \subset I$, then $\bigcap_{i\in I} K_i \neq \emptyset$.

Proof. Fix i_0. If the intersection were empty, the open sets $\mathbb{R}^n - K_i$ would cover K_{i_0}, and finitely many $\mathbb{R}^n - K_{i_1}, \ldots, \mathbb{R}^n - K_{i_m}$ would cover it; then $K_{i_0} \cap K_{i_1} \cap \cdots \cap K_{i_m} = \emptyset$, a contradiction. ■

In particular a decreasing sequence of nonempty compact sets has nonempty intersection.

Corollary 2.27 (2.5.6)

A nonempty countable closed set $S \subset \mathbb{R}^n$ has an isolated point.

Proof. Let $S = \{s_1, s_2, \ldots\}$ and suppose no point is isolated. Choose a closed ball $K_1 = \overline{N(y_1, r_1)}$ with $y_1 \in S$; inductively, since y_k is not isolated, $N(y_k, r_k)$ contains infinitely many points of S, so we can choose $y_{k+1} \in S \cap N(y_k, r_k)$, $y_{k+1} \neq s_k$, and a closed ball K_{k+1} about y_{k+1} inside $N(y_k, r_k)$ not containing s_k. The sets $K_k \cap S$ are nonempty, compact and decreasing, so they have a common point (Theorem 2.26); it lies in S but differs from every s_k, a contradiction. ■

Thus a nonempty closed set without isolated points (a **perfect set**), such as the Cantor set, is uncountable.

2.6 Connected sets

Motivation. Intervals are the sets that cannot be split into two separated pieces. This property is what lies behind the intermediate value theorem.

$S \subset \mathbb{R}^n$ is **disconnected** if there are open U, V with

$$U \cap S \neq \emptyset, \quad V \cap S \neq \emptyset, \quad U \cap V \cap S = \emptyset, \quad S \subset U \cup V,$$

and **connected** otherwise. A set $S \subset \mathbb{R}$ is an **interval** if $x, y \in S$ and $x < z < y$ imply $z \in S$.

Proposition 2.28 (2.6.1)

A subset S of $\mathbb{R}$ is connected iff it is an interval.

Proof. If S is not an interval, take $x < z < y$ with $x, y \in S$, $z \notin S$; then $U = (-\infty, z)$, $V = (z, \infty)$ disconnect S. Conversely let S be an interval and U, V as above, with $x \in U \cap S$, $y \in V \cap S$, say $x < y$; then $[x, y] \subset S$. Let $c = \sup([x, y] \cap U)$. If $c \in U$, then $c < y$ (as $y \notin U$) and U contains $[c, c+\delta)$, contradicting the definition of c; if $c \in V$, then $c > x$ and $(c-\delta, c] \subset V$, so $(c-\delta, c]$ misses U and c is not the supremum. But $c \in [x, y] \subset S \subset U \cup V$, a contradiction. ■

Proposition 2.29 (2.6.2)

If $\{C_i\}$ are connected and $\bigcap C_i \neq \emptyset$, then $C = \bigcup C_i$ is connected.

Proof. Suppose U, V disconnect C, and let $p \in \bigcap C_i$, say $p \in U$. Each C_i is connected and meets U (at p), so $C_i \cap V = \emptyset$ (otherwise U, V would disconnect C_i). Then $C \cap V = \emptyset$, a contradiction. ■

Convex sets, in particular balls and boxes, are connected: they are unions of segments through a common point, each segment being connected as a continuous image of an interval (Theorem 3.13).

2.7 Selected exercises

Motivation. Three problems from the book's exercise set: the diameter gives a convenient reformulation of the Cauchy condition and of the nested set theorems, and two short arguments show how rigid connectedness is.

For a nonempty $X \subset \mathbb{R}^n$ let $\operatorname{diam} X = \sup\{\|x - y\| : x, y \in X\} \in [0, \infty]$.

Exercise 2.7.23 (Diameter)

(a) Show that $\operatorname{diam} \overline{X} = \operatorname{diam} X$ for every nonempty $X \subset \mathbb{R}^n$.
(b) Show that a sequence (x_k) in $\mathbb{R}^n$ is Cauchy iff $\lim_{N\to\infty} \operatorname{diam}\{x_k : k \geq N\} = 0$.
(c) Let $X_1 \supset X_2 \supset \cdots$ be nonempty closed bounded sets with $\operatorname{diam} X_k \to 0$. Show that $\bigcap_k X_k$ consists of exactly one point.

Solution.

(a)

Step 1. $X \subset \overline{X}$ gives $\operatorname{diam} X \leq \operatorname{diam} \overline{X}$. We may assume $\operatorname{diam} X < \infty$.

Step 2. Let $x', y' \in \overline{X}$ and $\epsilon > 0$. By Proposition 2.3(b) there are $x, y \in X$ with $\|x - x'\| < \epsilon$ and $\|y - y'\| < \epsilon$. Then

$$\|x' - y'\| \leq \|x' - x\| + \|x - y\| + \|y - y'\| < \operatorname{diam} X + 2\epsilon.$$

As ϵ is arbitrary, $\|x' - y'\| \le \operatorname{diam} X$, and taking the supremum, $\operatorname{diam} \overline{X} \le \operatorname{diam} X$.

(b) Let $d_N = \operatorname{diam}\{x_k : k \ge N\}$; it is decreasing in N, since the sets shrink.

Step 1. If (x_k) is Cauchy and $\epsilon > 0$, choose N with $\|x_m - x_k\| < \epsilon/2$ for $m, k \ge N$. Then $d_M \le d_N \le \epsilon/2 < \epsilon$ for all $M \ge N$, so $d_N \to 0$.

Step 2. Conversely, if $d_N \to 0$ and $\epsilon > 0$, choose N with $d_N < \epsilon$; for $m, k \ge N$, $\|x_m - x_k\| \le d_N < \epsilon$.

(c)

Step 1. Each X_k is compact by the Heine–Borel theorem (Theorem 2.25), and the X_k are nonempty and decreasing, so $\bigcap_k X_k \neq \emptyset$ by Theorem 2.26.

Step 2. If $x, y \in \bigcap_k X_k$, then $\|x - y\| \le \operatorname{diam} X_k$ for every k, so $\|x - y\| = 0$ and $x = y$.

The hypothesis $\operatorname{diam} X_k \to 0$ is needed for uniqueness ($X_k = [0, 1]$), and closedness for existence ($X_k = (0, 1/k)$). ■

Exercise 2.7.25 (Sets between a connected set and its closure)

Let $A \subset \mathbb{R}^n$ be connected and $A \subset B \subset \overline{A}$. Show that B is connected. In particular $\overline{A}$ is connected.

Solution.

Step 1. Suppose open U, V disconnect B: $U \cap B \neq \emptyset$, $V \cap B \neq \emptyset$, $U \cap V \cap B = \emptyset$, $B \subset U \cup V$.

Step 2. Since $A \subset B$, we get $U \cap V \cap A = \emptyset$ and $A \subset U \cup V$. As A is connected, U and V cannot both meet A; say $U \cap A = \emptyset$.

Step 3. Pick $x \in U \cap B$. Since U is open, $N(x, \epsilon) \subset U$ for some $\epsilon > 0$. Since $x \in B \subset \overline{A}$, Proposition 2.3(b) gives a point of A in $N(x, \epsilon) \subset U$, contradicting $U \cap A = \emptyset$. Hence B is connected. ■

Exercise 2.7.26 (Connected sets have no isolated points)

Let $A \subset \mathbb{R}^n$ be connected with at least two points. Show that every point of A is a limit point of A.

Solution.

Step 1. Suppose $x \in A$ is not a limit point of A: there is $\epsilon > 0$ with $N(x, 2\epsilon) \cap A = \{x\}$. Let

$$U = N(x, \epsilon), \qquad V = \{y \in \mathbb{R}^n : \|y - x\| > \epsilon\}.$$

U is open, and V is open because $y \mapsto \|y - x\|$ is continuous (or directly: if $\|y - x\| > \epsilon$, then $N(y, \|y - x\| - \epsilon) \subset V$ by the triangle inequality). Also $U \cap V = \emptyset$.

Step 2. $x \in U \cap A$. Let $z \in A$, $z \neq x$ (it exists by assumption). Then $z \notin N(x, 2\epsilon)$, so $\|z - x\| \ge 2\epsilon > \epsilon$ and $z \in V \cap A$.

Step 3. Every $z \in A$ lies in $U \cup V$: either $z = x \in U$, or $\|z - x\| \ge 2\epsilon$ and $z \in V$. So U, V disconnect A, a contradiction. ■

Chapter 3. Continuous Functions

Throughout, $X \subset \mathbb{R}^n$, $Y \subset \mathbb{R}^m$, and functions $f : X \to Y$.

3.1 Limits and continuity

Motivation. Continuity means small changes in input produce small changes in output. We define it via limits and via sequences, and show the two agree, so that the sequence results of Chapters 1–2 apply directly to functions.

Let x_0 be a limit point of X. We write $\lim_{x\to x_0} f(x) = y_0$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - y_0\| < \epsilon$ whenever $x \in X$ and $0 < \|x - x_0\| < \delta$. f is **continuous at** $x_0 \in X$ if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(x_0)\| < \epsilon$ whenever $x \in X$, $\|x - x_0\| < \delta$; equivalently, x_0 is isolated in X, or $\lim_{x\to x_0} f(x) = f(x_0)$. f is **continuous** if it is continuous at every point.

Proposition 3.1 (3.1.1)

If $\lim_{x\to x_0} f(x) = y_0$ and $\lim_{x\to x_0} g(x) = z_0$, then $f + g \to y_0 + z_0$, $af \to ay_0$, and for real-valued functions $fg \to y_0 z_0$, and $f/g \to y_0/z_0$ if $z_0 \neq 0$.

Proof. As in Theorem 1.14, or by Proposition 3.3 below applied to sequences. ■

Proposition 3.2 (3.1.2)

If $f, g : X \to \mathbb{R}^m$ are continuous, so is $af + bg$; for real-valued f, g, so are fg and f/g (where $g \neq 0$).

Proof. Apply Proposition 3.1 at each non-isolated point. ■

Polynomials in several variables are therefore continuous, and so are rational functions where the denominator does not vanish.

Proposition 3.3 (3.1.3, sequential criterion)

f is continuous at x_0 iff $f(x_k) \to f(x_0)$ for every sequence $x_k \to x_0$ in X.

Proof. ($\Rightarrow$) Given ϵ, take δ; for large k, $\|x_k - x_0\| < \delta$, so $\|f(x_k) - f(x_0)\| < \epsilon$.

($\Leftarrow$) If f is not continuous at x_0, for some ϵ and each k there is x_k with $\|x_k - x_0\| < 1/k$ but $\|f(x_k) - f(x_0)\| \geq \epsilon$. ■

Similarly $\lim_{x\to x_0} f(x) = y_0$ iff $f(x_k) \to y_0$ for every sequence $x_k \to x_0$ with $x_k \neq x_0$.

Proposition 3.4 (3.1.4)

If $f : X \to Y$ is continuous at x_0 and $g : Y \to Z$ is continuous at $f(x_0)$, then $g \circ f$ is continuous at x_0.

Proof. Given ϵ, choose η for g at $f(x_0)$ and then δ for f at x_0 with η in place of ϵ. ■

Example 3.1

(a) $f(x) = \sin(1/x)$ $(x \neq 0)$ has no limit at 0: $x_k = 1/(k\pi + \pi/2)$ gives ± 1 alternately.
(b) $x \sin(1/x) \to 0$ by the squeeze $|x \sin(1/x)| \leq |x|$.
(c) Thomae's function ($f(p/q) = 1/q$ in lowest terms, $f = 0$ at irrationals) is continuous exactly at the irrationals.
(d) $f(x, y) = xy/(x^2 + y^2)$, $f(0, 0) = 0$, is continuous along every line through the origin, but $f(t, t) = \frac{1}{2}$, so f is not continuous at 0.

Theorem 3.5 (3.1.5)

For $f : X \to Y$ the following are equivalent:
(a) f is continuous;
(b) $f^{-1}(V)$ is open in X for every open V in Y;
(c) $f^{-1}(E)$ is closed in X for every closed E in Y.

Proof. (a)⇒(b): if $x \in f^{-1}(V)$ and $N(f(x), \epsilon) \cap Y \subset V$, the δ of continuity gives $N(x, \delta) \cap X \subset f^{-1}(V)$. (b)⇒(a): $f^{-1}(N(f(x_0), \epsilon) \cap Y)$ is open in X and contains x_0. (b)⇔(c): $f^{-1}(Y - E) = X - f^{-1}(E)$. ■

3.2 Continuity, compactness and connectedness

Motivation. The classical theorems about continuous functions on $[a, b]$, that they attain a maximum and take every intermediate value, are really statements that continuity preserves compactness and connectedness. Proving them this way makes them hold in $\mathbb{R}^n$ as well.

Theorem 3.6 (3.2.1)

If $f : X \to Y$ is continuous and surjective and X is compact, then Y is compact.

Proof. Let $\{U_i\}$ be an open cover of Y; by Theorem 3.5 and Proposition 2.6, $f^{-1}(U_i) = V_i \cap X$ with V_i open in $\mathbb{R}^n$, and the V_i cover X. Finitely many $V_{i_1}, \ldots, V_{i_k}$ do; then $U_{i_1}, \ldots, U_{i_k}$ cover $Y = f(X)$. ■

Corollary 3.7 (3.2.2, extreme value theorem)

A continuous $f : X \to \mathbb{R}$ on a compact set attains its maximum and minimum.

Proof. $f(X)$ is compact, hence bounded and closed; so $\sup f(X)$ is finite and, being a limit of points of $f(X)$, lies in $f(X)$. ■

Corollary 3.8 (3.2.3)

If X is compact and $f : X \to \mathbb{R}$ is continuous with $f > 0$, then there is $\delta > 0$ with $f(x) \geq \delta$ for all x.

Proof. Take $\delta = \min f > 0$. ■

Corollary 3.9 (3.2.4)

A continuous $f : [a, b] \to \mathbb{R}$ attains its maximum and minimum.

Corollary 3.10 (3.2.5)

If X is compact and $f : X \to Y$ is a continuous bijection, then f^{-1} is continuous.

Proof. Let U be open in X. $X - U$ is closed in the compact X, so compact (Proposition 2.24); then $f(X - U) = Y - f(U)$ is compact, hence closed, so $f(U)$ is open in Y. By Theorem 3.5, f^{-1} is continuous. ■

Compactness is needed: $t \mapsto (\cos t, \sin t)$ is a continuous bijection $[0, 2\pi) \to S^1$ whose inverse is not continuous. For nonempty $A, B \subset \mathbb{R}^n$ let $\operatorname{dist}(A, B) = \inf\{\|a - b\| : a \in A, b \in B\}$.

Proposition 3.11 (3.2.6)

If A is compact, B is closed and $A \cap B = \emptyset$, then $\operatorname{dist}(A, B) > 0$.

Proof. The function $d(x) = \operatorname{dist}(\{x\}, B)$ is continuous (indeed $|d(x) - d(y)| \le \|x - y\|$) and positive on A, since $\mathbb{R}^n - B$ is open. By Corollary 3.8 it is bounded below on A by some $\delta > 0$. ■

The compactness is needed: the closed sets $\{(x, 0)\}$ and $\{(x, 1/x) : x > 0\}$ are at distance 0.

Lemma 3.12 (3.2.7)

S is disconnected iff there is a continuous surjection $f : S \to \{0, 1\}$.

Proof. If U, V disconnect S, let $f = 0$ on $U \cap S$ and 1 on $V \cap S$; preimages of subsets of $\{0, 1\}$ are relatively open, so f is continuous. Conversely, $f^{-1}(0)$ and $f^{-1}(1)$ are nonempty and relatively open (Theorem 3.5), so they are $U \cap S$ and $V \cap S$ with U, V open (Proposition 2.6); then $U \cap V \cap S = f^{-1}(0) \cap f^{-1}(1) = \emptyset$ and $S \subset U \cup V$. ■

Theorem 3.13 (3.2.8)

If $f : X \to Y$ is continuous and surjective and X is connected, then Y is connected.

Proof. If Y were disconnected, Lemma 3.12 gives a continuous surjection $g : Y \to \{0, 1\}$, and then $g \circ f : X \to \{0, 1\}$ is a continuous surjection. ■

Corollary 3.14 (3.2.9)

If $I \subset \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$ is continuous, then $f(I)$ is an interval.

Proof. Proposition 2.28 and Theorem 3.13. ■

Corollary 3.15 (3.2.10, intermediate value theorem)

If $f : [a, b] \to \mathbb{R}$ is continuous and $f(a) < \alpha < f(b)$, then $f(c) = \alpha$ for some $c \in (a, b)$.

Corollary 3.16 (3.2.11)

If $f : [a, b] \to \mathbb{R}$ is continuous, then $f([a, b])$ is a bounded closed interval $[m, M]$.

Proof. It is an interval (Corollary 3.14) and compact (Theorem 3.6), containing its minimum m and maximum M. ■

A function mapping intervals to intervals need not be continuous, for example $\sin(1/x)$ extended by 0 at 0.

Corollary 3.17 (3.2.12, fixed point)

Every continuous $f : [a, b] \to [a, b]$ has a fixed point.

Proof. $g(x) = x - f(x)$ has $g(a) \le 0 \le g(b)$; apply Corollary 3.15 (or $g(a) = 0$ or $g(b) = 0$). ■

Example 3.2

Every polynomial of odd degree has a real root (it takes both signs). Every $a > 0$ has a unique positive n-th root, since x^n is continuous, strictly increasing and unbounded on $[0, \infty)$.

3.3 Uniform continuity

Motivation. Ordinary continuity allows the required δ to depend on the point. When one δ works everywhere we can approximate a function uniformly by step functions or polynomials, which is what integration needs; on compact sets this always happens.

$f : X \to Y$ is **uniformly continuous** if for every $\epsilon > 0$ there is $\delta > 0$ with $\|f(x) - f(y)\| < \epsilon$ whenever $x, y \in X$, $\|x - y\| < \delta$.

Theorem 3.18 (3.3.1, Heine)

A continuous function on a compact set is uniformly continuous.

Proof. Given ϵ, choose for each $x \in X$ a $\delta(x) > 0$ with $\|f(y)-f(x)\| < \epsilon/2$ for $y \in X$, $\|y-x\| < \delta(x)$. The sets $N(x, \delta(x)/2)$ cover X; finitely many with centres $x_1, \dots, x_k$ do. Let $\delta = \min \delta(x_i)/2$. If $\|x - y\| < \delta$, then $\|x - x_i\| < \delta(x_i)/2$ for some i, and $\|y - x_i\| < \delta(x_i)$; so $\|f(x) - f(y)\| \le \|f(x) - f(x_i)\| + \|f(x_i) - f(y)\| < \epsilon$. ■

Example 3.3

x^2 is not uniformly continuous on $\mathbb{R}$ ($(n+1/n)^2 - n^2 > 2$), and $1/x$ is not uniformly continuous on $(0, 1)$; $\sqrt{x}$ is uniformly continuous on $[0, \infty)$ since $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x - y|}$.

Theorem 3.19 (3.3.2)

A uniformly continuous function maps Cauchy sequences to Cauchy sequences.

Proof. Given ϵ, take δ from uniform continuity and N with $\|x_m - x_n\| < \delta$ for $m, n \ge N$. ■

Corollary 3.20 (3.3.3)

A uniformly continuous $f : X \to \mathbb{R}^m$ extends uniquely to a continuous function on $\overline{X}$, which is again uniformly continuous.

Proof. For $x_0 \in \overline{X}$ pick $x_k \to x_0$ in X; $(f(x_k))$ is Cauchy (Theorem 3.19), hence converges (Theorem 2.14); define $F(x_0)$ as the limit. Interlacing two such sequences shows the limit does not depend on the sequence, and $F = f$ on X. If $\|x - y\| < \delta$ in $\overline{X}$, approximate by $x_k \to x$, $y_k \to y$ with $\|x_k - y_k\| < \delta$; then $\|F(x) - F(y)\| = \lim \|f(x_k) - f(y_k)\| \le \epsilon$. Uniqueness follows from Proposition 3.3. ■

3.4 Monotone functions

Motivation. Monotone functions are the simplest functions that need not be continuous. Their discontinuities are only jumps and are countable, which makes them well suited to integration and to describing distribution functions.

For $f : (a, b) \to \mathbb{R}$ write $f(x+) = \lim_{t \to x^+} f(t)$ and $f(x-) = \lim_{t \to x^-} f(t)$ when these exist.

Theorem 3.21 (3.4.1)

A monotone $f : (a, b) \to \mathbb{R}$ has one-sided limits at every point, and $f(a+)$, $f(b-)$ exist in $[-\infty, \infty]$. If f is increasing and $a < x < y < b$, then $f(x-) \le f(x) \le f(x+) \le f(y-) \le f(y) \le f(y+)$, and $f(x+) = \inf_{t>x} f(t)$, $f(x-) = \sup_{t<x} f(t)$.

Proof. Let f increase. $A = \{f(t) : a < t < x\}$ is bounded above by $f(x)$; let $s = \sup A$. Given ϵ, some $f(t_0) > s - \epsilon$, and for $t_0 < t < x$, $s - \epsilon < f(t_0) \le f(t) \le s$; so $f(x-) = s \le f(x)$. Similarly $f(x+) = \inf_{t>x} f(t) \ge f(x)$. For $x < y$, choosing $x < t < y$, $f(x+) \le f(t) \le f(y-)$. ■

Corollary 3.22 (3.4.2)

The set of discontinuities of a monotone function on an interval is countable.

Proof. For increasing f, f is discontinuous at x iff $f(x-) < f(x+)$. Choose a rational $r(x) \in (f(x-), f(x+))$; by Theorem 3.21 these open intervals are disjoint for distinct x, so $x \mapsto r(x)$ is injective into $\mathbb{Q}$. ■

Conversely, for any countable set $\{x_k\} \subset (a, b)$, $f(x) = \sum_{x_k < x} 2^{-k}$ is increasing and discontinuous exactly at the x_k.

Theorem 3.23 (3.4.3)

For a continuous $f : I \to \mathbb{R}$ on an interval, f is injective iff it is strictly monotone. In that case $f(I)$ is an interval and $f^{-1} : f(I) \to I$ is continuous.

Proof. Strictly monotone functions are injective. Conversely, suppose f injective but not strictly monotone; then there are $x < y < z$ in I with $f(y)$ not between $f(x)$ and $f(z)$, say $f(y) > \max(f(x), f(z))$. A value α with $\max(f(x), f(z)) < \alpha < f(y)$ is attained in (x, y) and in (y, z) (Corollary 3.15), contradicting injectivity. If f is strictly increasing, f^{-1} is strictly increasing on the interval $f(I)$ (Corollary 3.14) and maps it onto the interval I; a monotone function with an interval as image has no jumps (a jump $g(y-) < g(y+)$ would omit values by Theorem 3.21), so f^{-1} is continuous. ■

3.5 Selected exercises

Motivation. Four problems from the book's exercise set. Each turns one tool of this chapter (density plus the sequential criterion, compactness, connectedness, completeness) into a concrete statement about continuous functions.

By Proposition 2.3(b) and Theorem 2.4, a set $F \subset \mathbb{R}^n$ is closed iff the limit of every convergent sequence in F lies in F; we use this repeatedly.

Exercise 3.5.2 (Cauchy's functional equation)

Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $f(x+y) = f(x) + f(y)$ for all x, y. Show that $f(x) = ax$ for all x, where $a = f(1)$.

Solution.

Step 1. $f(0) = f(0+0) = 2f(0)$, so $f(0) = 0$; then $0 = f(r + (-r)) = f(r) + f(-r)$ gives $f(-r) = -f(r)$.

Step 2. By induction on $n \in \mathbb{N}$, $f(nr) = nf(r)$: it holds for $n = 0$, and $f((n+1)r) = f(nr)+f(r) = (n+1)f(r)$. With Step 1 it holds for all $n \in \mathbb{Z}$.

Step 3. For $n \geq 1$, $a = f(1) = f(n \cdot \frac{1}{n}) = nf(\frac{1}{n})$, so $f(\frac{1}{n}) = \frac{a}{n}$, and for $m \in \mathbb{Z}$, $f(\frac{m}{n}) = mf(\frac{1}{n}) = a\frac{m}{n}$. Thus $f(q) = aq$ for every $q \in \mathbb{Q}$.

Step 4. Let $x \in \mathbb{R}$. By Theorem 1.8 choose $q_k \in \mathbb{Q}$ with $x < q_k < x + 1/k$, so $q_k \to x$. By Proposition 3.3, $f(x) = \lim f(q_k) = \lim aq_k = ax$. ■

Exercise 3.5.4 (Closed graphs)

For $f : \mathbb{R} \to \mathbb{R}$ let $G = \{(x, f(x)) : x \in \mathbb{R}\} \subset \mathbb{R}^2$ be its graph.
(a) Show that if f is continuous, then G is closed.
(b) Show that if f is bounded and G is closed, then f is continuous.
(c) Show by an example that boundedness cannot be dropped in (b).

Solution.

(a) Let $(x_k, f(x_k)) \to (x, y)$. Then $x_k \to x$ and $f(x_k) \to y$ (coordinatewise convergence). By Proposition 3.3, $f(x_k) \to f(x)$, so $y = f(x)$ and $(x, y) \in G$. Hence G is closed.

(b)

Step 1. Let $|f| \leq M$ and suppose f is not continuous at x. By Proposition 3.3 there are $\epsilon > 0$ and $x_k \to x$ with $|f(x_k) - f(x)| \geq \epsilon$ for all k.

Step 2. $(f(x_k))$ is bounded, so a subsequence $f(x_{k_j})$ converges to some y (Theorem 2.10), and $|y - f(x)| \geq \epsilon$.

Step 3. Then $(x_{k_j}, f(x_{k_j})) \to (x, y)$, and since G is closed, $(x, y) \in G$, i.e. $y = f(x)$. This contradicts Step 2.

(c) Let $f(x) = 1/x$ for $x \neq 0$ and $f(0) = 0$; f is not continuous at 0. Its graph is closed: let $(x_k, f(x_k)) \to (x, y)$.

Step 1. If $x \neq 0$, then $x_k \neq 0$ for large k and $f(x_k) = 1/x_k \to 1/x$, so $y = f(x)$.

Step 2. If $x = 0$: the convergent sequence $(f(x_k))$ is bounded, say by $K > 0$. Whenever $x_k \neq 0$, $1/|x_k| = |f(x_k)| \leq K$, i.e. $|x_k| \geq 1/K$. Since $x_k \to 0$, this forces $x_k = 0$ for all large k, and $y = \lim f(x_k) = 0 = f(0)$. ■

Exercise 3.5.9 (Antipodal points with equal values)

Let $T = \{(\cos t, \sin t) : t \in \mathbb{R}\} \subset \mathbb{R}^2$ be the unit circle and $f : T \to \mathbb{R}$ continuous. Show that $f(x) = f(-x)$ for some $x \in T$.

Solution.

Step 1. T is the image of the interval $[0, 2\pi]$ under the continuous map $t \mapsto (\cos t, \sin t)$, so T is connected by Proposition 2.28 and Theorem 3.13.

Step 2. $x \mapsto -x$ maps T into T and is continuous, so $g(x) = f(x) - f(-x)$ is continuous on T (Propositions 3.2 and 3.4). By Theorem 3.13 and Proposition 2.28, $g(T)$ is an interval.

Step 3. With $p = (1, 0)$ we have $g(-p) = f(-p) - f(p) = -g(p)$. So the interval $g(T)$ contains $g(p)$ and $-g(p)$, hence 0. A point x with $g(x) = 0$ satisfies $f(x) = f(-x)$.

For example, at any moment there are two antipodal points on the equator with the same temperature. ■

Exercise 3.5.10 (Contraction mapping principle)

Let $X \subset \mathbb{R}^n$ be nonempty and closed, $0 < r < 1$, and $f : X \to X$ with $\|f(x)-f(y)\| \le r\|x-y\|$ for all $x, y \in X$.
(a) Show that f is uniformly continuous.
(b) For $x_0 \in X$ let $x_{k+1} = f(x_k)$. Show that (x_k) is a Cauchy sequence.
(c) Show that f has exactly one fixed point x, that $x_k \to x$, and that $\|x_k - x\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Solution.

(a) Given $\epsilon > 0$ take $\delta = \epsilon$: if $\|x - y\| < \delta$, then $\|f(x) - f(y)\| \le r\|x - y\| < \epsilon$.

(b)

Step 1. By induction, $\|x_{j+1} - x_j\| = \|f(x_j) - f(x_{j-1})\| \le r\|x_j - x_{j-1}\| \le \cdots \le r^j\|x_1 - x_0\|$.

Step 2. For $m > k$, by the triangle inequality,

$$\|x_m - x_k\| \le \sum_{j=k}^{m-1} \|x_{j+1} - x_j\| \le \|x_1 - x_0\| \sum_{j=k}^{\infty} r^j = \frac{r^k}{1 - r}\|x_1 - x_0\|.$$

Step 3. Since $r^k \to 0$ (Example 1.1(b)), the right side is $< \epsilon$ for all large k, so (x_k) is Cauchy.

(c)

Step 1. By Theorem 2.14, $x_k \to x$ for some $x \in \mathbb{R}^n$, and $x \in X$ because X is closed.

Step 2. f is continuous by (a), so $f(x) = \lim f(x_k) = \lim x_{k+1} = x$ (Proposition 3.3).

Step 3. If also $f(y) = y$, then $\|x - y\| = \|f(x) - f(y)\| \le r\|x - y\|$, so $(1 - r)\|x - y\| \le 0$ and $y = x$.

Step 4. Letting $m \to \infty$ in Step 2 of (b) gives the error bound $\|x - x_k\| \le \frac{r^k}{1-r}\|x_1 - x_0\|$.

Closedness is needed: $f(x) = x/2$ on $X = (0, 1]$ is a contraction without a fixed point. ■

Chapter 4. Differentiable Functions

4.1 Differentiability

Motivation. The derivative is the best linear approximation to a function near a point. Its algebraic rules (product, quotient, chain) follow from writing $f(x+h) = f(x) + f'(x)h + o(h)$.

Let I be an interval. $f : I \to \mathbb{R}$ is **differentiable** at $x_0 \in I$ if $f'(x_0) = \lim_{h\to 0} \frac{f(x_0+h)-f(x_0)}{h}$ exists (one-sided at an end point).

Proposition 4.1 (4.1.1)

For $f : (a, b) \to \mathbb{R}$ and $x_0 \in (a, b)$ the following are equivalent:
(a) f is differentiable at x_0;
(b) there are $\lambda \in \mathbb{R}$ and a function ϕ with $f(x_0+h)-f(x_0) = \lambda h+|h|\phi(h)$ and $\lim_{h\to 0} \phi(h) = 0$.
Then $\lambda = f'(x_0)$.

Proof. (a)⇒(b): put $\lambda = f'(x_0)$ and $\phi(h) = (f(x_0 + h) - f(x_0) - \lambda h)/|h|$ for $h \neq 0$, $\phi(0) = 0$.
(b)⇒(a): $\frac{f(x_0+h)-f(x_0)}{h} = \lambda \pm \phi(h) \to \lambda$. ■

The linear map $h \mapsto \lambda h$ is the best linear approximation of $f(x_0 + h) - f(x_0)$; this is the form of the derivative that generalizes to several variables.

Proposition 4.2 (4.1.2)

If f is differentiable at x_0, it is continuous at x_0.

Proof. $f(x_0 + h) - f(x_0) = \lambda h + |h|\phi(h) \to 0$. ■

Proposition 4.3 (4.1.3)

If f, g are differentiable at x_0, so are $f+g$, fg, and f/g (if $g(x_0) \neq 0$), with $(f+g)' = f'+g'$, $(fg)' = f'g + fg'$, $(f/g)' = (f'g - fg')/g^2$ at x_0.

Proof. For the product, $\frac{f(x_0+h)g(x_0+h)-f(x_0)g(x_0)}{h} = \frac{f(x_0+h)-f(x_0)}{h} g(x_0+h) + f(x_0)\frac{g(x_0+h)-g(x_0)}{h}$, and $g(x_0 + h) \to g(x_0)$ by Proposition 4.2. For the quotient, first $\frac{1/g(x_0+h)-1/g(x_0)}{h} = -\frac{g(x_0+h)-g(x_0)}{h} \cdot \frac{1}{g(x_0+h)g(x_0)} \to -g'/g^2$, then use the product rule. ■

Proposition 4.4 (4.1.4, chain rule)

If $f : I \to J$ is differentiable at x_0 and $g : J \to \mathbb{R}$ is differentiable at $y_0 = f(x_0)$, then $g \circ f$ is differentiable at x_0 and $(g \circ f)'(x_0) = g'(y_0)f'(x_0)$.

Proof. Write $g(y_0 + k) - g(y_0) = k(g'(y_0) + \psi(k))$ with $\psi(k) \to 0 = \psi(0)$. With $k = f(x_0 + h) - f(x_0) \to 0$,

$$\frac{g(f(x_0 + h)) - g(f(x_0))}{h} = \frac{k}{h}(g'(y_0) + \psi(k)) \to f'(x_0)g'(y_0).$$

■

Example 4.1

$f(x) = x^2 \sin(1/x)$, $f(0) = 0$, is differentiable everywhere with $f'(0) = \lim h \sin(1/h) = 0$, but $f'(x) = 2x \sin(1/x) - \cos(1/x)$ is not continuous at 0. $|x|$ is continuous but not differentiable at 0.

4.2 Mean value theorems

Motivation. A derivative is local information, but we usually want global conclusions: $f' \geq 0$ implies f is increasing, $f' = 0$ implies f is constant. The mean value theorem converts one into the other, and its generalizations give L'Hôpital's rule.

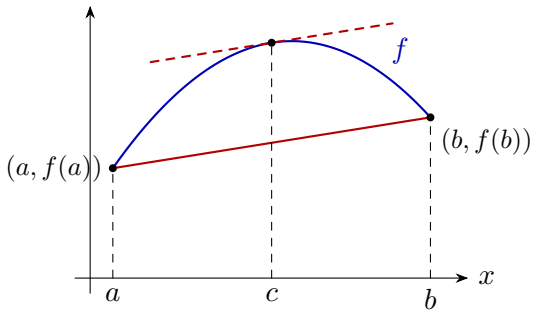


Figure. *The mean value theorem: at some $c \in (a, b)$ the tangent line (dashed) is parallel to the secant, i.e. $f'(c) = (f(b) - f(a))/(b - a)$.*

If f has a local maximum or minimum at an interior point c and is differentiable there, then $f'(c) = 0$: the difference quotients have opposite signs on the two sides.

Lemma 4.5 (4.2.1, Rolle)

If $f : [a, b] \to \mathbb{R}$ is continuous on $[a, b]$, differentiable on (a, b), and $f(a) = f(b)$, then $f'(c) = 0$ for some $c \in (a, b)$.

Proof. If f is constant, any c works. Otherwise f attains a maximum or minimum (Corollary 3.9) at a point $c \in (a, b)$ where $f(c) \neq f(a)$, and $f'(c) = 0$. ■

Theorem 4.6 (4.2.2, Cauchy mean value theorem)

If $f, g : [a, b] \to \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), there is $c \in (a, b)$ with $f'(c)[g(b) - g(a)] = g'(c)[f(b) - f(a)]$.

Proof. Apply Rolle to $h(x) = f(x)[g(b) - g(a)] - g(x)[f(b) - f(a)]$, for which $h(a) = h(b) = f(a)g(b) - g(a)f(b)$. ■

Corollary 4.7 (4.2.3, mean value theorem)

If f is continuous on $[a, b]$ and differentiable on (a, b), then $f(b) - f(a) = f'(c)(b - a)$ for some $c \in (a, b)$.

Proof. Take $g(x) = x$. ■

Corollary 4.8 (4.2.4)

For differentiable $f : (a, b) \to \mathbb{R}$: f is increasing iff $f' \geq 0$; constant iff $f' = 0$; decreasing iff $f' \leq 0$. If $f' > 0$, f is strictly increasing.

Proof. If $f' \geq 0$ and $x < y$, $f(y) - f(x) = f'(c)(y - x) \geq 0$. Conversely, if f increases, its difference quotients are ≥ 0, so $f' \geq 0$. ■

Theorem 4.9 (4.2.5, L'Hôpital's rule)

Let f, g be differentiable near x_0 (except possibly at x_0) with $g' \neq 0$ there, $\lim_{x\to x_0} f(x) = \lim_{x\to x_0} g(x) = 0$, and $\lim_{x\to x_0} f'(x)/g'(x) = L$. Then $\lim_{x\to x_0} f(x)/g(x) = L$. The same holds for one-sided limits, for $x \to \pm\infty$, and when $g(x) \to \pm\infty$.

Proof. Set $f(x_0) = g(x_0) = 0$, making f, g continuous at x_0. For x near x_0, $g(x) \neq 0$ (by Rolle, as $g' \neq 0$), and Theorem 4.6 on the interval between x_0 and x gives $f(x)/g(x) = f'(c_x)/g'(c_x)$ with c_x between x_0 and x; as $x \to x_0$, $c_x \to x_0$. For $x \to \infty$, substitute $x = 1/t$. If $g \to \infty$ as $x \to x_0^+$: given ϵ, choose δ with $|f'/g' - L| < \epsilon$ on $(x_0, x_0 + \delta)$ and fix y there. For $x_0 < x < y$, $\frac{f(x)-f(y)}{g(x)-g(y)} = \frac{f'(c)}{g'(c)}$, and

$$\frac{f(x)}{g(x)} = \frac{f(x) - f(y)}{g(x) - g(y)} \cdot \frac{1 - g(y)/g(x)}{1} + \frac{f(y)}{g(x)},$$

where $g(y)/g(x) \to 0$ and $f(y)/g(x) \to 0$ as $x \to x_0^+$; so $\limsup|f/g - L| \leq \epsilon$. ■

Theorem 4.10 (4.2.6, Darboux)

If $F : [a, b] \to \mathbb{R}$ is differentiable and $F'(a) < \alpha < F'(b)$, then $F'(c) = \alpha$ for some $c \in (a, b)$.

Proof. $G(x) = F(x) - \alpha x$ has $G'(a) < 0 < G'(b)$, so $G(x) < G(a)$ for x slightly larger than a and $G(x) < G(b)$ for x slightly smaller than b. Hence the minimum of G on $[a, b]$ is attained at an interior point c, where $G'(c) = 0$. ■

Corollary 4.11 (4.2.7)

If $f : I \to \mathbb{R}$ has a primitive on the interval I, then $f(J)$ is an interval for every interval $J \subset I$.

Proof. Apply Theorem 4.10 to the primitive on subintervals of J. ■

So a function with a jump discontinuity, such as the sign function, has no primitive; derivatives can be discontinuous (Example 4.1) but have the intermediate value property.

4.3 Taylor's theorem

Motivation. Polynomials are the functions we can compute with. Taylor's theorem says how well a smooth function is approximated by its Taylor polynomial and gives an explicit error term, which is the basis of local extremum tests and numerical approximations.

For f with n derivatives at a, let $T_n^f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x - a)^k$ be its Taylor polynomial at a.

Theorem 4.12 (4.3.1, generalized mean value theorem)

Let I be the closed interval with end points $a \neq b$, and $f, g : I \to \mathbb{R}$ with $f^{(n)}, g^{(n)}$ continuous on I and differentiable on $\operatorname{Int} I$. For each $x \in I$, $x \neq a$, there is c strictly between a and x with

$$[f(x) - T_n^f(x)]g^{(n+1)}(c) = [g(x) - T_n^g(x)]f^{(n+1)}(c).$$

Proof. Fix x and let $F(t) = \sum_{k=0}^{n} \frac{f^{(k)}(t)}{k!}(x-t)^k$, $G(t)$ likewise for g, for t between a and x. In $F'(t)$ the terms telescope:

$$F'(t) = \sum_{k=0}^{n} \frac{f^{(k+1)}(t)}{k!}(x-t)^k - \sum_{k=1}^{n} \frac{f^{(k)}(t)}{(k-1)!}(x-t)^{k-1} = \frac{f^{(n+1)}(t)}{n!}(x-t)^n,$$

and similarly for G. Also $F(x) = f(x)$ and $F(a) = T_n^f(x)$. Theorem 4.6 gives c with $[F(x) - F(a)]G'(c) = [G(x) - G(a)]F'(c)$; cancel the nonzero factor $(x-c)^n/n!$. ∎

Corollary 4.13 (4.3.2, Taylor's theorem)

Under the hypotheses on f, for each $x \in I$ there is c between a and x with

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1}.$$

Proof. Apply Theorem 4.12 with $g(t) = (t-a)^{n+1}$: $T_n^g = 0$ and $g^{(n+1)} = (n+1)!$. ∎

The last term is the **remainder**. If f is C^∞, the **Taylor series** of f at a is $\sum_k \frac{f^{(k)}(a)}{k!}(x-a)^k$; it may fail to converge to f: $f(x) = e^{-1/x^2}$, $f(0) = 0$, has all derivatives 0 at 0.

Proposition 4.14 (4.3.3)

Let f be C^∞ on I, $a, x \in I$, and J the closed interval between them. If $\{|f^{(n)}(t)| : t \in J, n \geq 0\}$ is bounded, the Taylor series of f at a converges to $f(x)$.

Proof. With M a bound, the remainder is at most $M|x-a|^{n+1}/(n+1)! \to 0$ (the terms of the convergent series $\sum r^n/n!$ tend to 0). ∎

Example 4.2

e^x, $\sin x$, $\cos x$ satisfy the hypothesis on every bounded interval, so $e^x = \sum x^n/n!$, $\sin x = \sum(-1)^n x^{2n+1}/(2n+1)!$, $\cos x = \sum(-1)^n x^{2n}/(2n)!$ for all real x. Taylor's theorem also gives the second derivative test: if $f'(a) = 0$ and $f''(a) > 0$, f has a strict local minimum at a.

4.4 Selected exercises

Motivation. Four problems from the book's exercise set. They show the mean value theorem, Rolle's lemma and L'Hôpital's rule at work on questions that are not immediate from the definitions.

Exercise 4.4.8 (A Hölder condition of order > 1)

Let I be an interval, $M > 0$, $\alpha > 1$, and $f : I \to \mathbb{R}$ with $|f(x) - f(y)| \le M|x - y|^\alpha$ for all $x, y \in I$. Show that f is constant. (The book's case is $M = 1$, $\alpha = 2$, $I = \mathbb{R}$.)

Solution.

Step 1. For $x \in I$ and $y \in I$, $y \neq x$,

$$\left|\frac{f(y) - f(x)}{y - x}\right| \le M|y - x|^{\alpha-1} \longrightarrow 0 \quad (y \to x),$$

since $\alpha - 1 > 0$. So f is differentiable on I with $f' \equiv 0$ (one-sided at end points).

Step 2. Let $a < b$ in I. f is continuous on $[a, b]$ (Proposition 4.2) and differentiable on (a, b), so by Corollary 4.7, $f(b) - f(a) = f'(c)(b - a) = 0$. Hence f is constant.

For $\alpha = 1$ the conclusion fails ($f(x) = x$), and for $\alpha < 1$ as well ($f(x) = \sqrt{x}$ on $[0, \infty)$ satisfies the condition with $\alpha = \frac{1}{2}$, $M = 1$, by Example 3.3). ■

Exercise 4.4.9 (Limits of the derivative)

Let I be an open interval, $c \in I$, and $f : I \to \mathbb{R}$ continuous at c and differentiable on $I - \{c\}$. Show that if $\lim_{x\to c} f'(x) = K$ exists, then f is differentiable at c and $f'(c) = K$.

Solution.

Step 1. Let $\epsilon > 0$. Choose $\eta > 0$ with $[c - \eta, c + \eta] \subset I$ and $|f'(z) - K| < \epsilon$ for $0 < |z - c| < \eta$.

Step 2. Let $0 < h < \eta$. f is continuous on $[c, c+h]$ (at c by hypothesis, elsewhere by Proposition 4.2) and differentiable on $(c, c + h)$. By Corollary 4.7 there is $z \in (c, c + h)$ with

$$\frac{f(c + h) - f(c)}{h} = f'(z), \qquad \text{so} \qquad \left|\frac{f(c + h) - f(c)}{h} - K\right| < \epsilon.$$

Step 3. The same argument on $[c + h, c]$ for $-\eta < h < 0$ gives the same bound. Hence the difference quotient tends to K, i.e. $f'(c) = K$.

Consequently a derivative has no removable or jump discontinuities: if f' has one-sided limits at c, they equal $f'(c)$. A derivative can still be discontinuous, as Example 4.1 shows, but only because $\lim_{x\to c} f'(x)$ fails to exist. ■

Exercise 4.4.12 (Roots of $\frac{d^n}{dx^n}(x^2 - x)^n$)

Let $n \ge 1$ and $f(x) = (x^2 - x)^n = x^n(x - 1)^n$. Show that $f^{(n)}$ has exactly n distinct roots, all in $(0, 1)$.

Solution.

Step 1. For $0 \le k \le n$, $f^{(k)}(x) = x^{n-k}h_k(x)$ for some polynomial h_k. This holds for $k = 0$ with $h_0 = (x - 1)^n$; if it holds for $k < n$, then with $m = n - k \ge 1$,

$$f^{(k+1)}(x) = mx^{m-1}h_k(x) + x^m h_k'(x) = x^{m-1}(mh_k(x) + xh_k'(x)).$$

Hence $f^{(k)}(0) = 0$ for $0 \le k \le n - 1$. Writing $f(x) = (x - 1)^n x^n$, the same argument with $x - 1$ in place of x gives $f^{(k)}(1) = 0$ for $0 \le k \le n - 1$.

Step 2. Claim: for $1 \le k \le n$, $f^{(k)}$ has at least k distinct roots in $(0, 1)$. For $k = 1$ apply Rolle's lemma (Lemma 4.5) to f on $[0, 1]$. If $k \le n - 1$ and $0 < \alpha_1 < \cdots < \alpha_k < 1$ are roots of $f^{(k)}$, then

by Step 1 $f^{(k)}$ vanishes at the $k+2$ points

$$0 < \alpha_1 < \cdots < \alpha_k < 1.$$

Rolle's lemma on each of the $k+1$ intervals between consecutive points gives a root of $f^{(k+1)}$ inside each; these open intervals are disjoint, so the $k+1$ roots are distinct.

Step 3. f has degree $2n$ with leading coefficient 1, so $f^{(n)}$ has degree n with leading coefficient $(2n)!/n! \neq 0$. A nonzero polynomial of degree n has at most n roots, so by Step 2 $f^{(n)}$ has exactly n distinct roots, all in $(0,1)$.

Up to a constant and the change of variable $x = (1+t)/2$, $f^{(n)}$ is the n-th Legendre polynomial. ■

Exercise 4.4.13 (Symmetric second difference quotient)

(a) Let f be differentiable on an open interval $I \ni c$ and suppose $f''(c)$ exists. Show that

$$\lim_{h\to 0} \frac{f(c+h) + f(c-h) - 2f(c)}{h^2} = f''(c).$$

(b) Show that the limit in (a) may exist although $f''(c)$ does not.

Solution.

(a)

Step 1. For small $h \neq 0$ let $N(h) = f(c+h) + f(c-h) - 2f(c)$ and $D(h) = h^2$. Both are differentiable in h, $N(h) \to 0$ by continuity of f at c, $D(h) \to 0$, and $D'(h) = 2h \neq 0$.

Step 2. We have

$$\frac{N'(h)}{D'(h)} = \frac{f'(c+h) - f'(c-h)}{2h} = \frac{1}{2}\Big(\frac{f'(c+h) - f'(c)}{h} + \frac{f'(c-h) - f'(c)}{-h}\Big)$$
$$\longrightarrow \frac{1}{2}\big(f''(c) + f''(c)\big) = f''(c).$$

Step 3. By L'Hôpital's rule (Theorem 4.9), $N(h)/D(h) \to f''(c)$.

(b) Let $f(x) = x|x|$ and $c = 0$. Then $N(h) = h|h| - h|h| - 0 = 0$, so the limit is 0. But $f'(x) = 2|x|$ for all x (at 0: $f(h)/h = |h| \to 0$), and $2|x|$ is not differentiable at 0; so $f''(0)$ does not exist. ■

Chapter 5. The Riemann–Stieltjes Integral

5.1 The Riemann integral

Motivation. Area under a graph is approximated by rectangles. The Riemann integral makes this rigorous by squeezing the true area between upper and lower sums; the work is in deciding which functions allow the squeeze.

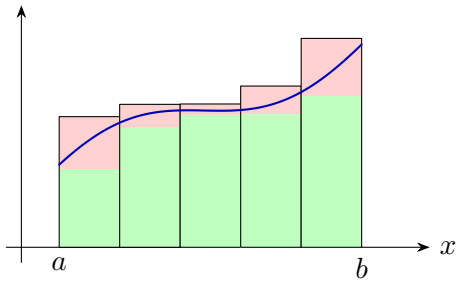


Figure. *Lower sum (green) and upper sum (green plus red) for a partition of $[a,b]$. f is integrable when refining the partition can make the red excess $U(f,P)-L(f,P)$ as small as we like.*

A **partition** of $[a,b]$ is a finite set $P=\{a=x_0<x_1<\cdots<x_n=b\}$; write $\Delta x_i=x_i-x_{i-1}$ and $\|P\|=\max \Delta x_i$ (the mesh). $Q\supset P$ is a **refinement** of P. For bounded $f:[a,b]\to\mathbb{R}$ let $M_i=\sup_{[x_{i-1},x_i]} f$, $m_i=\inf_{[x_{i-1},x_i]} f$, and

$$U(f,P)=\sum M_i\Delta x_i, \qquad L(f,P)=\sum m_i\Delta x_i.$$

The **upper** and **lower integrals** are $\overline{\int} f=\inf_P U(f,P)$ and $\underline{\int} f=\sup_P L(f,P)$; f is **Riemann integrable** ($f\in\mathcal{R}$) if they are equal, and the common value is $\int_a^b f$.

Proposition 5.1 (5.1.1)

If $P\subset Q$, then $L(f,P)\le L(f,Q)\le U(f,Q)\le U(f,P)$. For any partitions P,P', $L(f,P)\le U(f,P')$; hence $\underline{\int} f\le\overline{\int} f$.

Proof. It suffices to add one point $y\in(x_{i-1},x_i)$: the infima of f over $[x_{i-1},y]$ and $[y,x_i]$ are $\ge m_i$, so L can only increase; similarly U decreases. Then $L(f,P)\le L(f,P\cup P')\le U(f,P\cup P')\le U(f,P')$. ∎

Proposition 5.2 (5.1.2, Riemann's criterion)

A bounded f is integrable iff for every $\epsilon>0$ there is a partition P with $U(f,P)-L(f,P)<\epsilon$.

Proof. If the condition holds, $0\le\overline{\int} f-\underline{\int} f\le U(f,P)-L(f,P)<\epsilon$. Conversely, if $f\in\mathcal{R}$, choose P_1,P_2 with $U(f,P_1)<\int f+\epsilon/2$ and $L(f,P_2)>\int f-\epsilon/2$; then $P=P_1\cup P_2$ works by Proposition 5.1. ∎

Theorem 5.3 (5.1.3)

If $f,g\in\mathcal{R}$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}$ with $\int(f+g)=\int f+\int g$ and $\int cf=c\int f$; and $f\le g$ implies $\int f\le\int g$.

Proof. On each subinterval $\sup(f+g) \le \sup f + \sup g$ and $\inf(f+g) \ge \inf f + \inf g$, so

$$L(f,P) + L(g,P) \le L(f+g,P) \le U(f+g,P) \le U(f,P) + U(g,P).$$

Choosing P with $U - L < \epsilon/2$ for both f and g (common refinement) gives integrability and the formula. For $c \ge 0$, $U(cf,P) = cU(f,P)$ and $L(cf,P) = cL(f,P)$; for $c = -1$, $U(-f,P) = -L(f,P)$. If $f \le g$, then $U(f,P) \le U(g,P)$ for all P. ■

Products of integrable functions are integrable: $f^2 \in \mathcal{R}$ because on each subinterval $\sup f^2 - \inf f^2 \le 2K(M_i - m_i)$ with $K = \sup|f|$, and $fg = \frac{1}{4}[(f+g)^2 - (f-g)^2]$.

Theorem 5.4 (5.1.4)

If $f \in \mathcal{R}[a,b]$ and $a < c < b$, then $f \in \mathcal{R}[a,c]$, $f \in \mathcal{R}[c,b]$ and $\int_a^b f = \int_a^c f + \int_c^b f$. Conversely, integrability on $[a,c]$ and $[c,b]$ implies integrability on $[a,b]$.

Proof. For partitions P of $[a,b]$ containing c, with $P_1 = P \cap [a,c]$ and $P_2 = P \cap [c,b]$, we have $U(f,P) = U(f,P_1) + U(f,P_2)$, likewise for L. If $U(f,P) - L(f,P) < \epsilon$ (we may add c to P), the same holds for P_1 and P_2. Conversely, combine partitions of the two halves. Taking infima of upper sums gives the formula. ■

With the convention $\int_b^a f = -\int_a^b f$, the formula holds for any order of a, b, c.

Theorem 5.5 (5.1.5)

If $f \in \mathcal{R}$, then $|f| \in \mathcal{R}$ and $\left|\int_a^b f\right| \le \int_a^b |f|$.

Proof. From $||f(x)| - |f(y)|| \le |f(x) - f(y)|$, the oscillation of $|f|$ on each subinterval is at most that of f, so $U(|f|,P) - L(|f|,P) \le U(f,P) - L(f,P)$. Then $-|f| \le f \le |f|$ and Theorem 5.3 give the inequality. ■

Example 5.1

The Dirichlet function (1 on $\mathbb{Q}$, 0 elsewhere) is not integrable on $[0,1]$: every $U(f,P) = 1$ and every $L(f,P) = 0$. Integrability of $|f|$ does not imply that of f: take $f = 1$ on $\mathbb{Q}$ and -1 elsewhere.

5.2 Integrable functions

Motivation. Which functions are integrable? Continuous and monotone ones are, and integrability survives sums, products and composition with continuous functions. These results let us integrate everything we meet in practice without checking upper and lower sums each time.

Theorem 5.6 (5.2.1)

Every continuous $f : [a,b] \to \mathbb{R}$ is integrable.

Proof. f is uniformly continuous (Theorem 3.18): choose δ with $|f(x) - f(y)| < \epsilon/(b-a)$ when $|x-y| < \delta$. If $\|P\| < \delta$, then $M_i - m_i < \epsilon/(b-a)$ (the extrema are attained), so $U(f,P) - L(f,P) < \epsilon$. ■

Theorem 5.7 (5.2.2)

Every monotone $f:[a,b]\to\mathbb{R}$ is integrable.

Proof. Let f increase and P divide $[a,b]$ into n equal parts. Then $M_i=f(x_i)$, $m_i=f(x_{i-1})$, and $U(f,P)-L(f,P)=\frac{b-a}{n}\sum(f(x_i)-f(x_{i-1}))=\frac{b-a}{n}(f(b)-f(a))<\epsilon$ for large n. ∎

More generally, a bounded function with finitely many discontinuities is integrable (enclose them in short intervals), and so is one whose discontinuity set has measure zero (Chapter 10 of the second volume).

A **Riemann sum** is $S(f,P)=\sum f(t_i)\Delta x_i$ with $t_i\in[x_{i-1},x_i]$.

Theorem 5.8 (5.2.3)

For bounded f and $A\in\mathbb{R}$ the following are equivalent:
(a) $f\in\mathcal{R}$ and $\int_a^b f=A$;
(b) for every $\epsilon>0$ there is $\delta>0$ with $|S(f,P)-A|<\epsilon$ for every partition with $\|P\|<\delta$ and every choice of the t_i.

Proof. (a)⇒(b): let $K=\sup|f|$ and choose P_0, with k points, such that $U(f,P_0)-L(f,P_0)<\epsilon/2$. Let $\|P\|<\delta=\epsilon/(8Kk)$. Passing from P to $P\cup P_0$ changes only the at most k subintervals of P that contain points of P_0 in their interior, and each change of U or L is at most $2K\|P\|$. So $U(f,P)-L(f,P)\le U(f,P\cup P_0)-L(f,P\cup P_0)+4Kk\delta<\epsilon$. Since both $S(f,P)$ and A lie in $[L(f,P),U(f,P)]$, $|S(f,P)-A|<\epsilon$.

(b)⇒(a): fix P with $\|P\|<\delta$. Choosing the t_i with $f(t_i)>M_i-\epsilon/(b-a)$ gives $U(f,P)<S(f,P)+\epsilon<A+2\epsilon$; similarly $L(f,P)>A-2\epsilon$. So $U(f,P)-L(f,P)<4\epsilon$, and $\int f=A$. ∎

Example 5.2

$\int_0^1 x^2\,dx=\lim_n\frac{1}{n}\sum_{k=1}^n(k/n)^2=\lim\frac{(n+1)(2n+1)}{6n^2}=\frac{1}{3}$. Also $\lim_n\sum_{k=1}^n\frac{1}{n+k}=\int_0^1\frac{dx}{1+x}=\log 2$.

5.3 The fundamental theorem of calculus

Motivation. Differentiation and integration are inverse operations. This connects the area problem with antiderivatives and yields integration by parts and change of variables.

For $f\in\mathcal{R}[a,b]$ let $F(x)=\int_a^x f$.

Theorem 5.9 (5.3.1)

F is uniformly continuous. If f is continuous at x_0, then F is differentiable at x_0 and $F'(x_0)=f(x_0)$.

Proof. With $K=\sup|f|$, $|F(y)-F(x)|=|\int_x^y f|\le K|y-x|$. If f is continuous at x_0, given ϵ choose δ with $|f(t)-f(x_0)|<\epsilon$ for $|t-x_0|<\delta$; then for $0<|h|<\delta$,

$$\left|\frac{F(x_0+h)-F(x_0)}{h}-f(x_0)\right|=\left|\frac{1}{h}\int_{x_0}^{x_0+h}(f(t)-f(x_0))\,dt\right|\le\epsilon.$$

∎

Corollary 5.10 (5.3.2)

Every continuous $f : [a, b] \to \mathbb{R}$ has a primitive, namely $F(x) = \int_a^x f$.

Theorem 5.11 (5.3.3, fundamental theorem of calculus)

If $F : [a, b] \to \mathbb{R}$ is differentiable and $f = F'$ is integrable, then $\int_a^b f = F(b) - F(a)$.

Proof. For any partition, the mean value theorem gives $t_i \in (x_{i-1}, x_i)$ with $F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i$, so $F(b) - F(a) = \sum f(t_i)\Delta x_i \in [L(f, P), U(f, P)]$. As $f \in \mathcal{R}$, $F(b) - F(a) = \int f$. ∎

The integrability hypothesis cannot be dropped: there are differentiable functions with bounded, non-integrable derivatives (Volterra), and $x^2 \sin(1/x^2)$ has an unbounded derivative on $[0, 1]$.

Proposition 5.12 (5.3.4, change of variables)

If $g : [a, b] \to [c, d]$ is a surjective C^1 function and $f : [c, d] \to \mathbb{R}$ is continuous, then $\int_{g(a)}^{g(b)} f(x)\,dx = \int_a^b f(g(t))g'(t)\,dt$.

Proof. Let F be a primitive of f (Corollary 5.10). By the chain rule $(F \circ g)' = (f \circ g)g'$, which is continuous, hence integrable, so by Theorem 5.11 the right side is $F(g(b)) - F(g(a))$, which is also the left side. ∎

Proposition 5.13 (5.3.5, integration by parts)

If $F, G : [a, b] \to \mathbb{R}$ are differentiable and $f = F'$, $g = G'$ are integrable, then $\int_a^b Fg = F(b)G(b) - F(a)G(a) - \int_a^b fG$.

Proof. F, G are continuous, hence integrable, so $(FG)' = fG + Fg$ is integrable (products of integrable functions); apply Theorem 5.11 to FG. ∎

5.4 Functions of bounded variation

Motivation. Which functions are the difference of two increasing functions? Exactly those whose total up-and-down movement is finite. Such functions are the natural integrators in the Riemann–Stieltjes integral and describe rectifiable curves.

For $f : [a, b] \to \mathbb{R}$ and a partition P, let $V(f, P) = \sum |f(x_i) - f(x_{i-1})|$. The **total variation** is $V_a^b(f) = \sup_P V(f, P)$; f is of **bounded variation** if it is finite. Refining P increases $V(f, P)$ (triangle inequality). Monotone functions have $V_a^b(f) = |f(b) - f(a)|$; Lipschitz functions with constant K have $V_a^b(f) \le K(b - a)$; BV functions are bounded, $|f(x)| \le |f(a)| + V_a^b(f)$.

Proposition 5.14 (5.4.1)

If f, g are of bounded variation, then so are $f + g$ and fg, with

$$V(f + g) \le V(f) + V(g), \qquad V(fg) \le \|f\|_\infty V(g) + \|g\|_\infty V(f).$$

If moreover $|f| \ge c > 0$, then $1/f$ is of bounded variation with $V(1/f) \le V(f)/c^2$.

Proof. Termwise, $|(f+g)(x_i)-(f+g)(x_{i-1})| \le |\Delta f_i| + |\Delta g_i|$, and

$$|f(x_i)g(x_i) - f(x_{i-1})g(x_{i-1})| \le |f(x_i)||\Delta g_i| + |g(x_{i-1})||\Delta f_i|,$$

and $|1/f(x_i) - 1/f(x_{i-1})| = |\Delta f_i|/|f(x_i)f(x_{i-1})| \le |\Delta f_i|/c^2$. Sum and take suprema. ∎

Theorem 5.15 (5.4.2)

If f is of bounded variation on $[a,b]$ and $a<c<b$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_a^b(f) = V_a^c(f) + V_c^b(f)$. Conversely, bounded variation on both halves implies it on $[a,b]$.

Proof. For partitions $P \ni c$ of $[a,b]$, $V(f,P) = V(f, P\cap[a,c]) + V(f, P\cap[c,b])$, and every partition may be refined to contain c. Taking suprema gives the formula, both sides being finite or infinite together. ∎

Theorem 5.16 (5.4.3, Jordan decomposition)

$f:[a,b]\to\mathbb{R}$ is of bounded variation iff $f = g - h$ with g, h increasing.

Proof. Monotone functions are of bounded variation, hence so are their differences (Proposition 5.14). Conversely, let $g(x) = V_a^x(f)$ (with $g(a)=0$) and $h = g - f$. g increases by Theorem 5.15, and for $x<y$, $h(y)-h(x) = V_x^y(f) - (f(y)-f(x)) \ge 0$. ∎

Consequently a BV function has one-sided limits everywhere and at most countably many discontinuities (Theorem 3.21, Corollary 3.22). A continuous function need not be of bounded variation: $x\sin(1/x)$ on $[0,1]$.

A **curve** is a continuous $\alpha:[a,b]\to\mathbb{R}^m$; its **length** is $\Lambda(\alpha) = \sup_P \sum \|\alpha(x_i) - \alpha(x_{i-1})\|$.

Proposition 5.17 (5.4.4)

For a curve $\alpha = (\alpha_1,\dots,\alpha_m)$ the following are equivalent:
(a) $\Lambda(\alpha) < \infty$;
(b) each α_k is of bounded variation.

Proof. For each k, $|\alpha_k(x_i) - \alpha_k(x_{i-1})| \le \|\alpha(x_i) - \alpha(x_{i-1})\| \le \sum_j |\alpha_j(x_i) - \alpha_j(x_{i-1})|$. Summing, $V(\alpha_k, P) \le \sum_i \|\Delta\alpha_i\| \le \sum_j V(\alpha_j, P)$; take suprema. ∎

Theorem 5.18 (5.4.5)

Every C^1 curve $\alpha:[a,b]\to\mathbb{R}^m$ has finite length $\Lambda(\alpha) = \int_a^b \|\alpha'(t)\|\,dt$.

Proof. For vector-valued v (integrated componentwise), $\|\int v\| \le \int \|v\|$: with $u = \int v$, $\|u\|^2 = \int\langle u, v\rangle \le \|u\| \int\|v\|$ by Cauchy–Schwarz. Hence $\|\alpha(x_i) - \alpha(x_{i-1})\| = \|\int_{x_{i-1}}^{x_i} \alpha'\| \le \int_{x_{i-1}}^{x_i} \|\alpha'\|$, and $\Lambda(\alpha) \le \int\|\alpha'\|$. Conversely, α' is uniformly continuous; for $\|P\|$ small, $\|\alpha'(t) - \alpha'(x_i)\| < \epsilon$ on $[x_{i-1}, x_i]$, so

$$\int_{x_{i-1}}^{x_i} \|\alpha'\| \le \|\alpha'(x_i)\|\Delta x_i + \epsilon\Delta x_i = \Big\| \int_{x_{i-1}}^{x_i} (\alpha'(t) + \alpha'(x_i) - \alpha'(t))\,dt \Big\| + \epsilon\Delta x_i \le \|\Delta\alpha_i\| + 2\epsilon\Delta x_i.$$

Summing, $\int\|\alpha'\| \le \Lambda(\alpha) + 2\epsilon(b-a)$. ∎

5.5 The Riemann–Stieltjes integral

Motivation. Replacing lengths Δx by increments $\Delta\alpha$ of a monotone function lets one integral handle sums, ordinary integrals, and expectations of random variables with any distribution.

Let $\alpha:[a,b]\to\mathbb{R}$ be increasing. With $\Delta\alpha_i=\alpha(x_i)-\alpha(x_{i-1})$ define $U(f,P,\alpha)=\sum M_i\Delta\alpha_i$, $L(f,P,\alpha)=\sum m_i\Delta\alpha_i$, and $S(f,P,\alpha)=\sum f(t_i)\Delta\alpha_i$. As in Proposition 5.1, refinement decreases U and increases L. $f\in\mathcal{R}(\alpha)$ if $\inf_P U(f,P,\alpha)=\sup_P L(f,P,\alpha)$; the common value is $\int_a^b f\,d\alpha$.

Theorem 5.19 (5.5.1)

Let α,β be increasing and f,g bounded.
(a) If $f,g\in\mathcal{R}(\alpha)$ and $c\in\mathbb{R}$, then $f+g, cf\in\mathcal{R}(\alpha)$ with $\int(f+g)\,d\alpha=\int f\,d\alpha+\int g\,d\alpha$ and $\int cf\,d\alpha=c\int f\,d\alpha$; and $f\le g$ implies $\int f\,d\alpha\le\int g\,d\alpha$.
(b) If $f\in\mathcal{R}(\alpha)$ on $[a,b]$ and $a<c<b$, then $f\in\mathcal{R}(\alpha)$ on $[a,c]$ and $[c,b]$, and $\int_a^b=\int_a^c+\int_c^b$.
(c) If $f\in\mathcal{R}(\alpha)\cap\mathcal{R}(\beta)$ and $c\ge0$, then $f\in\mathcal{R}(\alpha+\beta)\cap\mathcal{R}(c\alpha)$ with $\int f\,d(\alpha+\beta)=\int f\,d\alpha+\int f\,d\beta$ and $\int f\,d(c\alpha)=c\int f\,d\alpha$.
(d) If $f\in\mathcal{R}(\alpha)$, then $|f|\in\mathcal{R}(\alpha)$ and $\left|\int f\,d\alpha\right|\le\int|f|\,d\alpha\le\sup|f|\,(\alpha(b)-\alpha(a))$.

Proof. The proofs of Theorems 5.3, 5.4 and 5.5 apply verbatim with Δx_i replaced by $\Delta\alpha_i\ge0$. For (c), $U(f,P,\alpha+\beta)=U(f,P,\alpha)+U(f,P,\beta)$ and likewise for L. ∎

Theorem 5.20 (5.5.2)

For bounded f and increasing α the following are equivalent:
(a) $f\in\mathcal{R}(\alpha)$;
(b) for every $\epsilon>0$ there is P with $U(f,P,\alpha)-L(f,P,\alpha)<\epsilon$;
(c) there is A such that for every $\epsilon>0$ some P_0 has $|S(f,P,\alpha)-A|<\epsilon$ for every $P\supset P_0$ and every choice of tags. In that case $A=\int f\,d\alpha$.

Proof. (a)⇔(b) as in Proposition 5.2.

(b)⇒(c): with $A=\int f\,d\alpha$ and P_0 as in (b), for $P\supset P_0$ both $S(f,P,\alpha)$ and A lie in $[L(f,P,\alpha),U(f,P,\alpha)]$, an interval of length $<\epsilon$.

(c)⇒(b): for $P\supset P_0$, taking suprema and infima over tags, $U(f,P,\alpha)\le A+\epsilon$ and $L(f,P,\alpha)\ge A-\epsilon$. ∎

Example 5.3

(a) If α is the unit step at $c\in(a,b)$ ($\alpha=0$ on $[a,c]$, 1 on $(c,b]$) and f is continuous at c, then $\int f\,d\alpha=f(c)$. If f and α are both discontinuous from the right at c, f may fail to be integrable.
(b) For a step function α with jumps c_k at points s_k and continuous f, $\int f\,d\alpha=\sum c_k f(s_k)$: sums are a special case of Stieltjes integrals.

For α of bounded variation, write $\alpha=\alpha_1-\alpha_2$ with α_i increasing (Theorem 5.16). If $f\in\mathcal{R}(\alpha_1)\cap\mathcal{R}(\alpha_2)$ for some such decomposition, we say $f\in\mathcal{R}(\alpha)$ and set $\int f\,d\alpha=\int f\,d\alpha_1-\int f\,d\alpha_2$; by Theorem 5.22 below this does not depend on the decomposition. The Riemann–Stieltjes sum $S(f,P,\alpha)$ is defined as before.

Proposition 5.21 (5.5.3)

If f is continuous on $[a,b]$, then $f\in\mathcal{R}(\alpha)$ for every α of bounded variation.

Proof. It suffices to treat increasing α. Given ϵ, by uniform continuity choose P with $M_i - m_i < \epsilon$ for all i; then $U(f,P,\alpha) - L(f,P,\alpha) \le \epsilon(\alpha(b)-\alpha(a))$. ∎

Theorem 5.22 (5.5.4)

For bounded f and α of bounded variation the following are equivalent:
(a) $f \in \mathcal{R}(\alpha)$ and $\int f\,d\alpha = A$;
(b) for every $\epsilon > 0$ there is P_0 with $|S(f,P,\alpha) - A| < \epsilon$ for all $P \supset P_0$ and all tags.

Proof. (a)⇒(b): $S(f,P,\alpha) = S(f,P,\alpha_1) - S(f,P,\alpha_2)$; apply Theorem 5.20(c) to each and take a common refinement. In particular A is determined by f and α, so the integral is well defined.

(b)⇒(a): let $V(x) = V_a^x(\alpha)$; we show $f \in \mathcal{R}(V)$. Let $K = \sup|f|$ and $\epsilon > 0$. Choose P_0 as in (b) and refine it so that $V(b) - \sum_i |\Delta\alpha_i| < \epsilon$ (definition of $V(b)$; refining keeps this). For $P \supset P_0$ write

$$U(f,P,V) - L(f,P,V) = \sum (M_i - m_i)(\Delta V_i - |\Delta\alpha_i|) + \sum (M_i - m_i)|\Delta\alpha_i|.$$

Since $\Delta V_i \ge |\Delta\alpha_i|$, the first sum is at most $2K\epsilon$. For the second, choose s_i, t_i in the i-th interval with $f(t_i) - f(s_i) > M_i - m_i - \epsilon$, and swap them where $\Delta\alpha_i < 0$; then the second sum is less than $\sum (f(t_i) - f(s_i))\Delta\alpha_i + \epsilon V(b) = S(f,P,\alpha) - S'(f,P,\alpha) + \epsilon V(b) < 2\epsilon + \epsilon V(b)$, where S, S' are sums with tags t_i, s_i. By Theorem 5.20, $f \in \mathcal{R}(V)$. Then the sums $S(f,P,V-\alpha) = S(f,P,V) - S(f,P,\alpha)$ satisfy the condition of Theorem 5.20(c) for the increasing function $V - \alpha$, so $f \in \mathcal{R}(V-\alpha)$, and with the decomposition $\alpha = V - (V-\alpha)$, $\int f\,d\alpha = A$. ∎

Theorem 5.23 (5.5.5)

Let α be differentiable with α' continuous on $[a,b]$. If f is bounded and $f \in \mathcal{R}(\alpha)$, then $f\alpha'$ is Riemann integrable and $\int_a^b f\,d\alpha = \int_a^b f(x)\alpha'(x)\,dx$.

Proof. α is of bounded variation (Lipschitz). By the mean value theorem $\Delta\alpha_i = \alpha'(u_i)\Delta x_i$ with $u_i \in (x_{i-1}, x_i)$, so

$$\left| S(f,P,\alpha) - \sum f(t_i)\alpha'(t_i)\Delta x_i \right| \le K \sum |\alpha'(u_i) - \alpha'(t_i)|\Delta x_i.$$

By uniform continuity of α' this is $\le K\epsilon(b-a)$ once $\|P\|$ is small. Combining with Theorem 5.22(b), the Riemann sums of $f\alpha'$ over all sufficiently fine refinements of some P_0 are within $\epsilon + K\epsilon(b-a)$ of $\int f\,d\alpha$. By Theorem 5.20 with $\alpha(x) = x$, $f\alpha' \in \mathcal{R}$ with the stated integral. ∎

Theorem 5.24 (5.5.6, integration by parts)

Let f, α be of bounded variation on $[a,b]$. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ and $\int_a^b f\,d\alpha + \int_a^b \alpha\,df = f(b)\alpha(b) - f(a)\alpha(a)$.

Proof. Let $A = \int f\,d\alpha$ and P_0 as in Theorem 5.22(b). Let $Q \supset P_0$ be a partition $a = y_0 < \cdots < y_m = b$ with tags $s_k \in [y_{k-1}, y_k]$. By summation by parts,

$$S(\alpha,Q,f) = \sum_k \alpha(s_k)(f(y_k) - f(y_{k-1})) = f(b)\alpha(b) - f(a)\alpha(a) - \sum_{k=0}^{m} f(y_k)(\alpha(s_{k+1}) - \alpha(s_k)),$$

with $s_0 = a$, $s_{m+1} = b$. The last sum is a Riemann–Stieltjes sum $S(f,P,\alpha)$ for the partition P formed by the points s_k together with the y_k (zero-length intervals contribute nothing), with tags y_k; $P \supset Q \supset P_0$. So $|S(\alpha,Q,f) - (f(b)\alpha(b) - f(a)\alpha(a) - A)| < \epsilon$, and Theorem 5.22 applies to α and f. ∎

Corollary 5.25 (5.5.7, second mean value theorem)

If $f : [a, b] \to \mathbb{R}$ is increasing and g is continuous, there is $c \in [a, b]$ with $\int_a^b fg = f(a) \int_a^c g + f(b) \int_c^b g$.

Proof. Let $G(x) = \int_a^x g$, a C^1 function with $G' = g$. $G \in \mathcal{R}(f)$ (Proposition 5.21), so $f \in \mathcal{R}(G)$ and $\int f\, dG = f(b)G(b) - \int G\, df$ (Theorem 5.24), while $\int f\, dG = \int fg$ (Theorem 5.23). With m, M the minimum and maximum of G, $m(f(b) - f(a)) \le \int G\, df \le M(f(b) - f(a))$, so $\int G\, df = G(c)(f(b) - f(a))$ for some c by the intermediate value theorem. Hence $\int fg = f(b)G(b) - G(c)(f(b) - f(a)) = f(a)G(c) + f(b)(G(b) - G(c))$. ■

Appendix A. The Number Systems

Chapter 1 took the real numbers as a complete ordered field. Here we construct such a field. We accept the arithmetic and order of $\mathbb{N} = \{0, 1, 2, \dots\}$, including the fact that every nonempty subset of $\mathbb{N}$ has a least element, build $\mathbb{Z}$ and $\mathbb{Q}$ by equivalence relations, and obtain $\mathbb{R}$ from Cauchy sequences of rationals.

A.1 Integers and rationals

Motivation. Chapter 1 took the real numbers as an axiomatic starting point. This appendix shows that such a field actually exists, by building it from the rationals; the first step is to recall how $\mathbb{Z}$ and $\mathbb{Q}$ themselves arise.

On $\mathbb{N} \times \mathbb{N}$ put $(m, n) \sim (m', n') \iff m + n' = n + m'$ (think of $m - n$); this is an equivalence relation, and $(m, n) \sim (m - k, n - k)$ when $m, n \geq k$. $\mathbb{Z}$ is the set of classes $[m, n]$, with $[m, n] \geq [k, l] \iff m + l \geq n + k$, $[m, n] + [k, l] = [m + k, n + l]$, $[m, n] \cdot [k, l] = [mk + nl, ml + nk]$. These are well defined (independent of representatives), and every integer is $[n, 0]$, $[0, 0]$ or $[0, n]$ for a unique $n \geq 1$, written n, 0, $-n$; the map $n \mapsto [n, 0]$ embeds $\mathbb{N}$ preserving operations and order.

Theorem A.1 (Theorem 1)

(a) For $a, b \in \mathbb{Z}$, $a \geq b$ or $b \geq a$.
(b) A nonempty subset of $\mathbb{Z}$ bounded above has a largest element, and one bounded below has a least element.

Proof. (a) reduces to the same property of $\mathbb{N}$ for $m + l$ and $n + k$.

(b) If S is bounded below by b, then $\{s - b : s \in S\} \subset \mathbb{N}$ has a least element k, and $b + k$ is least in S; for S bounded above apply this to $-S$. ∎

$\mathbb{Z}$ satisfies the field axioms except the existence of multiplicative inverses, and with $P_{\mathbb{Z}} = \{1, 2, \dots\}$ the order axioms. On $\mathbb{Z} \times (\mathbb{Z} - \{0\})$ put $(a, b) \sim (c, d) \iff ad = bc$ and order the classes by $(a, b) > (c, d) \iff abd^2 > cdb^2$; the classes are the rationals a/b, with $a/b + c/d = (ad + bc)/bd$ and $(a/b)(c/d) = ac/bd$. One checks that $\mathbb{Q}$ is an ordered field containing $\mathbb{Z}$, so the results of Section 1.1, valid in any ordered field, hold in $\mathbb{Q}$.

A.2 Real numbers as classes of Cauchy sequences

Motivation. A real number is what a Cauchy sequence of rationals "should" converge to. Taking equivalence classes of such sequences makes this literal and produces a complete ordered field, which justifies the axioms of Chapter 1.

A rational sequence $(a(i))$ is a **fundamental sequence** if for every rational $\epsilon > 0$ there is N with $|a(i) - a(j)| < \epsilon$ for $i, j \geq N$.

Theorem A.2 (Theorem 2)

(a) Fundamental sequences are bounded.
(b) A bounded monotone rational sequence is fundamental.
(c) If $(a(i))$ is fundamental, so is $(|a(i)|)$.
(d) If $(a(i)), (b(i))$ are fundamental, so are $(a(i) \pm b(i))$ and $(a(i)b(i))$.
(e) If $(a(i))$ is fundamental and there are N and a rational $d > 0$ with $|a(i)| \geq d$ for $i \geq N$,

then $(1/a(i))_{i\ge N}$ is fundamental.

Proof. (a) as in Proposition 2.12.

(b) If an increasing sequence bounded by B were not fundamental, there would be a rational $\epsilon>0$ and indices $i_1<j_1\le i_2<j_2\le\cdots$ with $a(j_k)-a(i_k)\ge\epsilon$; then $a(j_k)\ge a(i_1)+k\epsilon>B$ for large k.

(c) $||a(i)|-|a(j)||\le|a(i)-a(j)|$.

(d) $|a(i)b(i)-a(j)b(j)|\le|a(i)||b(i)-b(j)|+|b(j)||a(i)-a(j)|$ and (a).

(e) $|1/a(i)-1/a(j)|\le|a(i)-a(j)|/d^2$. ■

Two fundamental sequences are **equivalent**, $(a(i))\sim(b(i))$, if for every rational $\epsilon>0$ there is N with $|a(i)-b(i)|<\epsilon$ for $i\ge N$; this is an equivalence relation. A **real number** is an equivalence class $[a]$; $\mathbb{R}$ is the set of classes. Define $[a]>[b]$ if there are a rational $d>0$ and N with $a(i)-b(i)>d$ for $i\ge N$; this does not depend on the representatives.

Theorem A.3 (Theorem 3, trichotomy)

For $[a],[b]\in\mathbb{R}$ exactly one of $[a]>[b]$, $[a]=[b]$, $[b]>[a]$ holds.

Proof. For rational $\epsilon>0$ let N_ϵ be such that $|a(i)-a(N_\epsilon)|<\epsilon$ and $|b(i)-b(N_\epsilon)|<\epsilon$ for $i\ge N_\epsilon$. Then for $i\ge N_\epsilon$,

$$d_\epsilon:=a(N_\epsilon)-b(N_\epsilon)-2\epsilon<a(i)-b(i)<a(N_\epsilon)-b(N_\epsilon)+2\epsilon=:d'_\epsilon.$$

If $d_\epsilon>0$ for some ϵ, then $[a]>[b]$; if $d'_\epsilon<0$ for some ϵ, then $[b]>[a]$. Otherwise for every ϵ, $|a(N_\epsilon)-b(N_\epsilon)|\le2\epsilon$, so $|a(i)-b(i)|<4\epsilon$ for $i\ge N_\epsilon$, and $[a]=[b]$. Two of the cases cannot hold together, by the definitions. ■

Define $[a]+[b]=[a+b]$ and $[a][b]=[ab]$ (termwise operations); these are well defined by Theorem A.2. For $r\in\mathbb{Q}$ let r^* be the class of the constant sequence r; $r\mapsto r^*$ embeds $\mathbb{Q}$ in $\mathbb{R}$ preserving operations and order. 0^* and 1^* are the identities, $[-a]$ is the additive inverse, and if $[a]\ne0^*$ then by Theorem A.3 there are $d>0$ and N with $|a(i)|\ge d$ for $i\ge N$, so the class of $(1/a(i))_{i\ge N}$ (Theorem A.2(e)) is the inverse. With $P=\{[a]:[a]>0^*\}$, $\mathbb{R}$ is an ordered field, and $|[a]|=[|a|]$.

Theorem A.4 (Theorem 4, density of $\mathbb{Q}$)

If $[b]>[a]$, there is a rational r with $[b]>r^*>[a]$.

Proof. Choose N_1 and rational $d>0$ with $b(i)-a(i)>d$ for $i\ge N_1$, and N_2 with $|a(i)-a(j)|<d/4$, $|b(i)-b(j)|<d/4$ for $i,j\ge N_2$; let $N=\max(N_1,N_2)$ and $r=a(N)+d/2$. For $i\ge N$, $r-a(i)=d/2+(a(N)-a(i))>d/4$, so $r^*>[a]$; and $b(i)-r=(b(i)-b(N))+(b(N)-a(N))-d/2>-d/4+d-d/2=d/4$, so $[b]>r^*$. ■

Theorem A.5 (Theorem 5, completeness)

Every Cauchy sequence $([a_n])$ in $\mathbb{R}$ converges in $\mathbb{R}$.

Proof. If the sequence is eventually constant there is nothing to prove; otherwise, passing to a subsequence (which suffices by Lemma 2.13, whose proof is valid in any ordered field), assume $[a_n]\ne[a_{n+1}]$ for all n. By Theorem A.4 choose rationals $r(n)$ strictly between $[a_n]$ and $[a_{n+1}]$; then $|[a_n]-r(n)^*|<|[a_n]-[a_{n+1}]|\to0$. The rational sequence $(r(n))$ is fundamental, since $|r(n)-r(m)|^*\le|r(n)^*-[a_n]|+|[a_n]-[a_m]|+|[a_m]-r(m)^*|$. Let $[r]$ be its class. Given $[e]>0$ choose, by Theorem A.4, a rational $\epsilon>0$ with $\epsilon^*<[e]$, and M with $|r(i)-r(j)|<\epsilon/2$ for $i,j\ge M$.

For $n \geq M$, $|r(n)^* - [r]| \leq (\epsilon/2)^*$, because the termwise differences $r(n) - r(i)$ are eventually below $\epsilon/2$. Hence for large n, $|[a_n] - [r]| \leq |[a_n] - r(n)^*| + (\epsilon/2)^* < [e]$. ∎

Finally, Theorems A.4 and A.5 imply the completeness axiom: given S nonempty and bounded above, the bisection argument after Theorem 1.20 produces nested intervals with rational end points whose lengths tend to 0; the left end points form a Cauchy sequence, whose limit (Theorem A.5) is $\sup S$. So $\mathbb{R}$ is a complete ordered field, unique up to isomorphism by Theorem 1.10.

A.3 Selected exercises

Motivation. Two checks that the text states without proof: that the definitions on fundamental sequences do not depend on representatives, and that the new field contains a square root of 2, which $\mathbb{Q}$ does not.

Throughout, $a, b, \ldots$ denote fundamental sequences of rationals, and ϵ, d rational numbers.

Exercise A.1 (Well-definedness)

(a) Show that $\sim$ is an equivalence relation on fundamental sequences.
(b) Show that if $a \sim a'$ and $b \sim b'$, then $ab \sim a'b'$.
(c) Show that if $a \sim a'$, $b \sim b'$ and $[a] > [b]$ via a, b, then also $a'(i) - b'(i) > d'$ for some rational $d' > 0$ and all large i.

Solution [Supplement].

(a) Reflexivity and symmetry are clear from $|a(i) - a(i)| = 0$ and $|a(i) - b(i)| = |b(i) - a(i)|$. For transitivity, let $a \sim b$, $b \sim c$ and $\epsilon > 0$; choose N with $|a(i) - b(i)| < \epsilon/2$ and $|b(i) - c(i)| < \epsilon/2$ for $i \geq N$. Then $|a(i) - c(i)| < \epsilon$ for $i \geq N$.

(b)

Step 1. By Theorem A.2(a) there is a rational $B > 0$ with $|a(i)| \leq B$ and $|b'(i)| \leq B$ for all i. Then

$$|a(i)b(i) - a'(i)b'(i)| \leq |a(i)|\,|b(i) - b'(i)| + |b'(i)|\,|a(i) - a'(i)|.$$

Step 2. Given $\epsilon > 0$, choose N with $|b(i) - b'(i)| < \epsilon/(2B)$ and $|a(i) - a'(i)| < \epsilon/(2B)$ for $i \geq N$. Then the left side is $< \epsilon$ for $i \geq N$. Both products are fundamental by Theorem A.2(d), so $ab \sim a'b'$.

(c) Let $a(i) - b(i) > d$ for $i \geq N_1$. Choose N_2 with $|a(i) - a'(i)| < d/4$ and $|b(i) - b'(i)| < d/4$ for $i \geq N_2$. For $i \geq \max(N_1, N_2)$,

$$a'(i) - b'(i) \geq (a(i) - b(i)) - |a(i) - a'(i)| - |b(i) - b'(i)| > d - \tfrac{d}{2} = \tfrac{d}{2},$$

so $d' = d/2$ works. Hence the order on $\mathbb{R}$ does not depend on the representatives. ∎

Exercise A.2 (A square root of 2)

Define rationals $a(0) = 2$ and $a(i+1) = \frac{a(i)}{2} + \frac{1}{a(i)}$.
(a) Show that $a(i) > 0$, $a(i)^2 \geq 2$ and $a(i+1) \leq a(i)$ for all i; conclude that $(a(i))$ is fundamental.
(b) Show that $0 \leq a(i)^2 - 2 \leq 2 \cdot 4^{-i}$, and deduce that $x = [a]$ satisfies $x > 0^*$ and $x^2 = 2^*$.

Solution [Supplement].

(a)

Step 1. By induction $a(i)$ is a positive rational, since $a(i) > 0$ implies $a(i+1) > 0$.

Step 2. Expanding the square,

$$a(i+1)^2 - 2 = \frac{a(i)^2}{4} + 1 + \frac{1}{a(i)^2} - 2 = \Big(\frac{a(i)}{2} - \frac{1}{a(i)}\Big)^2 = \frac{\big(a(i)^2-2\big)^2}{4a(i)^2} \geq 0.$$

Together with $a(0)^2 = 4$, this gives $a(i)^2 \geq 2$ for all i.

Step 3. $a(i) - a(i+1) = \frac{a(i)}{2} - \frac{1}{a(i)} = \frac{a(i)^2-2}{2a(i)} \geq 0$. So $(a(i))$ is decreasing and bounded below by 1 (as $a(i) > 0$ and $a(i)^2 \geq 2 > 1$).

Step 4. Then $(-a(i))$ is increasing and bounded, hence fundamental by Theorem A.2(b), and so is $(a(i))$.

(b)

Step 1. Let $e(i) = a(i)^2 - 2 \geq 0$. By Step 2 of (a) and $a(i)^2 \geq 2$, $e(i+1) = e(i)^2/(4a(i)^2) \leq e(i)^2/8$.

Step 2. $e(0) = 2$. If $e(i) \leq 2 \cdot 4^{-i} \leq 2$, then $e(i+1) \leq e(i) \cdot \frac{e(i)}{8} \leq \frac{e(i)}{4} \leq 2 \cdot 4^{-(i+1)}$. By induction the bound holds for all i.

Step 3. Given a rational $\epsilon > 0$, $2 \cdot 4^{-i} < \epsilon$ for large i, so $|a(i)a(i) - 2| < \epsilon$ eventually: $aa \sim (2, 2, \dots)$, i.e. $x^2 = [aa] = 2^*$.

Step 4. $a(i) - 0 \geq 1 > \frac{1}{2}$ for all i, so $x > 0^*$ by the definition of the order (with $d = \frac{1}{2}$).

Since no rational has square 2, the fundamental sequence $(a(i))$ has no limit in $\mathbb{Q}$: this is exactly the incompleteness of $\mathbb{Q}$ that the construction repairs. ■